PROJECT UAS

"GlobalScale Retail Data Warehouse"

Disusun Guna Memenuhi Tugas Mata Kuliah

Data Warehouse

Dosen Pengampu :

Endah Septa Sintiya, S.Pd., M.Kom.

Disusun Oleh:

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Jurusan Teknologi Informasi
D4 Sistem Informasi Bisnis
Politeknik Negeri Malang

2026

I. STUDI KASUS DAN DATASET

A. Pemilihan Studi Kasus

Studi kasus yang dipilih dalam proyek ini adalah data penjualan ritel dari dataset publik yang tersedia di platform Kaggle dengan judul "Sample Sales Data". Dataset ini dipilih karena memenuhi seluruh kriteria yang dibutuhkan, yaitu kelengkapan atribut data, kemudahan akses, serta ketersediaan secara legal untuk keperluan akademik dan penelitian.

Link Dataset : https://www.kaggle.com/datasets/kyanyoga/sample-sales-data

B. Deskripsi Dataset

Tabel dan atribut yang ditemukan dalam dataset tersebut :

1. Tabel dimCustomer

Atribut	Keterangan
id_dimCustomer	Primary Key tabel dimensi pelanggan
customerName	Nama perusahaan/pelanggan
contactFirstName	Nama depan kontak pelanggan
contactLastName	Nama belakang kontak pelanggan
phone	Nomor telepon pelanggan

2. Tabel dimProducts

Atribut	Keterangan
id_dimProduct	Primary Key tabel dimensi produk
productCode	Kode unik produk
productLine	Kategori/lini produk
msrp	Manufacturer's Suggested Retail Price

3. **Tabel dimDate**

Atribut	Keterangan
id_dimDate	Primary Key tabel dimensi waktu
date	Tanggal transaksi (format lengkap)
month	Bulan transaksi
quarter	Kuartal transaksi
year	Tahun transaksi

4. **Tabel dimLocation**

Atribut	Keterangan
id_dimLocation	Primary Key tabel dimensi lokasi
addressLine1	Alamat baris pertama
addressLine2	Alamat baris kedua (opsional)
city	Kota
state	Provinsi/negara bagian
postalCode	Kode pos
country	Negara
territory	Wilayah/region

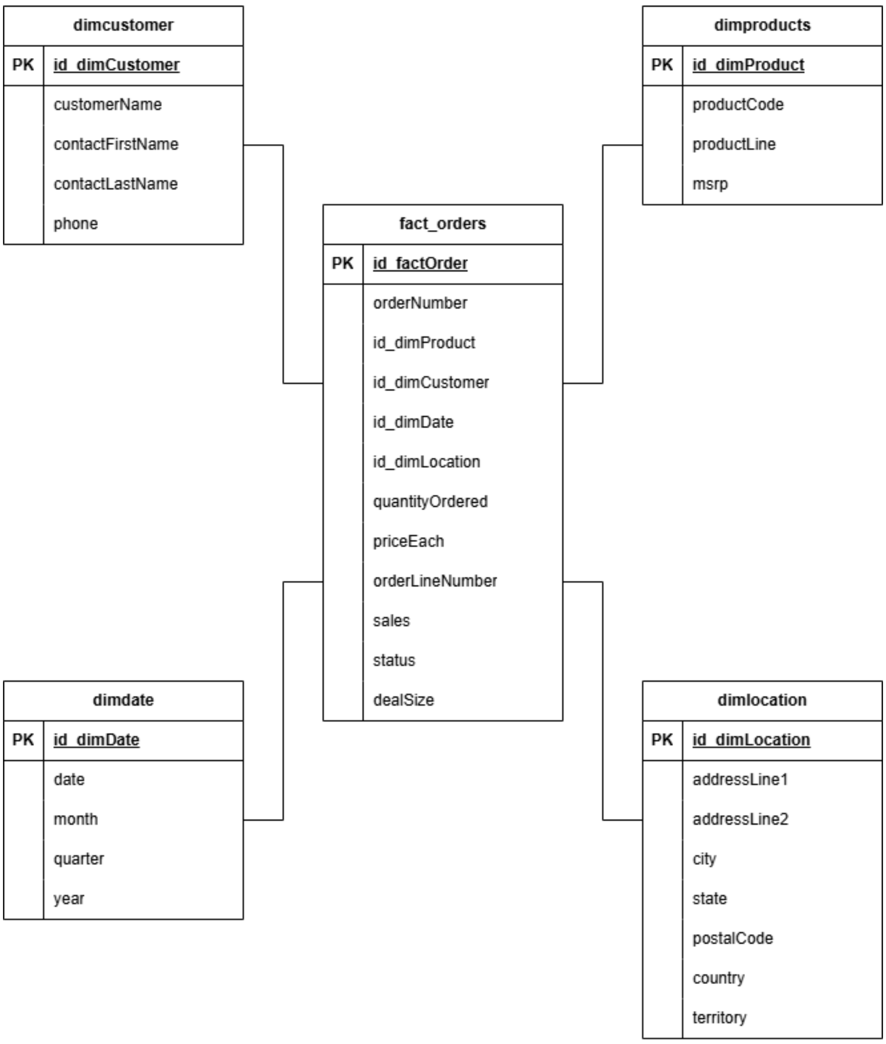
5. **Tabel fact_Orders (Tabel Fakta)**

Atribut	Keterangan
id_factOrder	Primary Key tabel fakta
orderNumber	Nomor pesanan
id_dimProduct (FK)	Foreign Key ke tabel dimProducts
id_dimCustomer (FK)	Foreign Key ke tabel dimCustomer

id_dimDate (FK)	Foreign Key ke tabel dimDate
id_dimLocation (FK)	Foreign Key ke tabel dimLocation
quantityOrdered	Jumlah item yang dipesan
priceEach	Harga satuan item
orderLineNumber	Nomor baris dalam pesanan
sales	Total nilai penjualan
status	Status pesanan (Shipped, Cancelled, dll.)
dealSize	Ukuran deal (Small, Medium, Large)

II. PERANCANGAN SKEMA BINTANG (STAR SCHEMA)

A. Konsep Star Schema



B. Tabel Fakta : fact_orders

Tabel fakta merupakan inti dari star schema yang menyimpan data numerik terukur (measures) serta foreign key yang menghubungkan ke seluruh tabel dimensi. Tabel fakta dalam proyek ini adalah fact_Orders yang merepresentasikan setiap baris transaksi penjualan.

Atribut ukuran (measures) dalam tabel fakta meliputi:

- quantityOrdered: jumlah produk yang dipesan dalam satu transaksi.
- priceEach: harga satuan produk pada saat transaksi.
- sales: total nilai penjualan (hasil perkalian quantityOrdered dan priceEach).

C. Tabel Dimensi

Tabel dimensi berfungsi sebagai konteks deskriptif bagi data fakta. Berikut adalah empat tabel dimensi yang dirancang :

1. **dimCustomer**

   Menyimpan informasi identitas pelanggan, termasuk nama perusahaan, nama kontak, dan nomor telepon. Dimensi ini memungkinkan analisis penjualan berdasarkan segmen pelanggan.

2. **dimProducts**

   Menyimpan informasi produk, termasuk kode produk, lini produk (kategori), dan harga MSRP. Dimensi ini memungkinkan analisis performa penjualan per produk atau per kategori.

3. **dimDate**

   Menyimpan atribut waktu yang diperlukan untuk analisis temporal, meliputi tanggal, bulan, kuartal, dan tahun. Dimensi waktu merupakan komponen kritis dalam setiap data warehouse karena hampir semua analisis bisnis memiliki dimensi waktu.

4. **dimLocation**

   Menyimpan informasi geografis pengiriman, meliputi alamat, kota, provinsi, kode pos, negara, dan wilayah/territory. Dimensi ini memungkinkan analisis penjualan berdasarkan wilayah geografis.

D. Relasi Antar Tabel

Relasi antar tabel dalam star schema GlobalScale Retail dibangun melalui primary key dan foreign key. Tabel fact_Orders sebagai tabel pusat menyimpan empat foreign key yang masing-masing merujuk ke id_dimCustomer, id_dimProduct, id_dimDate, dan id_dimLocation. Dengan struktur ini, setiap record transaksi dapat ditelusuri secara lengkap berdasarkan identitas pelanggan, produk yang dibeli, waktu transaksi, dan lokasi pengiriman.

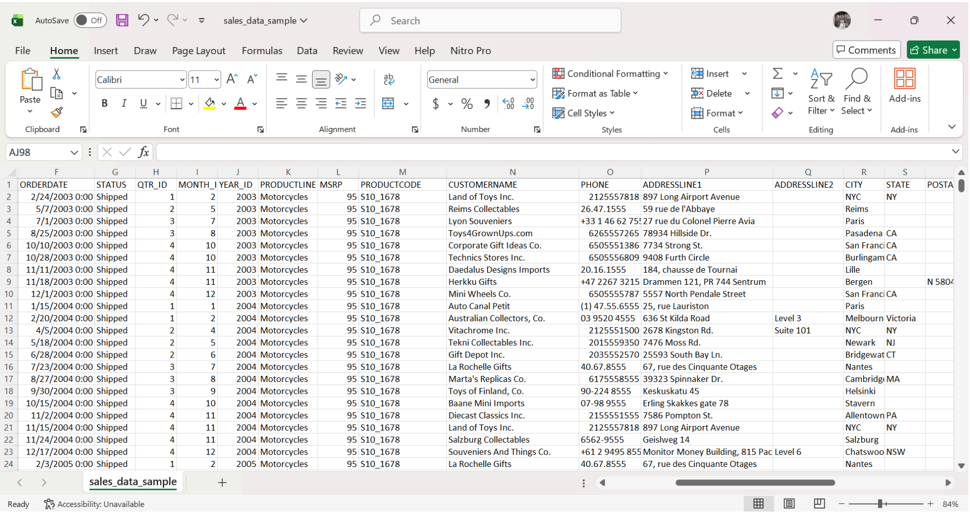
III. IMPLEMENTASI PIPELINE ETL

A. Gambaran Umum Pipeline ETL

ETL (Extract, Transform, Load) merupakan proses inti dalam pembangunan data warehouse. Pipeline ETL pada proyek GlobalScale Retail dirancang untuk mengalirkan data dari file CSV sumber ke dalam struktur data warehouse yang telah dirancang. Pipeline ini terdiri dari tiga tahapan utama yang dilakukan secara berurutan.

B. Tahap Extract (Ekstraksi Data)

1. Dataset sales_data_sample.csv dibuka di Excel, terdiri dari 2824 baris data dan 25 kolom yaitu ORDERNUMBER,QUANTITYORDERED, PRICEEACH,ORDERLINENUMBER,SALES,ORDERDATE, STATUS,QTR_ID,MONTH_ID,YEAR_ID,PRODUCTLINE, MSRP,PRODUCTCODE,CUSTOMERNAME,PHONE, ADDRESSLINE1,ADDRESSLINE2,CITY,STATE,POSTALCODE, COUNTRY,TERRITORY,CONTACTLASTNAME,CONTACTFIRSTNAME, DEALSIZE.



2. Identifikasi missing values

   Dari hasil inspeksi awal ditemukan 4 kolom yang memiliki nilai kosong yaitu ADDRESSLINE2 sebanyak 2521 baris, STATE sebanyak 1486 baris, POSTALCODE sebanyak 76 baris, dan TERRITORY sebanyak 1074 baris.

3. Identifikasi duplikasi

   Dari hasil pengecekan duplikasi tidak ditemukan baris data yang duplikat.

C. Inisialisasi Database & Skema DDL (Data Definition Language)

1. Membuat Database

```
CREATE DATABASE IF NOT EXISTS dw_sales;
USE dw_sales;
```

2. Membuat Tabel Dimensi (Dimension Tables)

```
-- Tabel Dimensi: dimcustomer
CREATE TABLE IF NOT EXISTS dimcustomer (
   id_dimCustomer   INT          NOT NULL AUTO_INCREMENT,
   customerName     VARCHAR(50)   NOT NULL,
   contactFirstName VARCHAR(50)   NOT NULL,
   contactLastName  VARCHAR(50)   NOT NULL,
   phone            VARCHAR(20)   NOT NULL,
   PRIMARY KEY (id_dimCustomer)
);

-- Tabel Dimensi: dimproducts
CREATE TABLE IF NOT EXISTS dimproducts (
   id_dimProduct  INT          NOT NULL AUTO_INCREMENT,
   productCode    VARCHAR(15)   NOT NULL,
   productLine    VARCHAR(50)   NOT NULL,
   msrp           DECIMAL(10,2) NOT NULL,
   PRIMARY KEY (id_dimProduct)
);
```

```
-- Tabel Dimensi: dimdate
CREATE TABLE IF NOT EXISTS dimdate (
   id_dimDate  INT  NOT NULL AUTO_INCREMENT,
   date        DATE NOT NULL,
   month       INT  NOT NULL,
   quarter     INT  NOT NULL,
   year        INT  NOT NULL,
   PRIMARY KEY (id_dimDate)
);

-- Tabel Dimensi: dimlocation
CREATE TABLE IF NOT EXISTS dimlocation (
   id_dimLocation  INT          NOT NULL AUTO_INCREMENT,
   addressLine1    VARCHAR(100)  NOT NULL,
   addressLine2    VARCHAR(100),
   city            VARCHAR(50)   NOT NULL,
   state           VARCHAR(50),
   postalCode      VARCHAR(15),
   country         VARCHAR(50)   NOT NULL,
   territory       VARCHAR(50),
   PRIMARY KEY (id_dimLocation)
);
```

3. **Membuat Tabel Fakta dengan Relasi Foreign Key (Fact Table)**

```
-- Tabel Fakta: fact_orders
CREATE TABLE IF NOT EXISTS fact_orders (
   id_factOrder    INT          NOT NULL AUTO_INCREMENT,
   orderNumber     INT          NOT NULL,
   id_dimProduct   INT          NOT NULL,
   id_dimCustomer  INT          NOT NULL,
   id_dimDate      INT          NOT NULL,
   id_dimLocation  INT          NOT NULL,
   quantityOrdered INT          NOT NULL,
   priceEach       DECIMAL(10,2) NOT NULL,
```

orderLineNumber INT NOT NULL,
sales DECIMAL(10,2) NOT NULL,
status VARCHAR(15) NOT NULL,
dealSize VARCHAR(10) NOT NULL,
PRIMARY KEY (id_factOrder),
FOREIGN KEY (id_dimProduct) REFERENCES dimproducts(id_dimProduct),
FOREIGN KEY (id_dimCustomer) REFERENCES dimcustomer(id_dimCustomer),
FOREIGN KEY (id_dimDate) REFERENCES dimdate(id_dimDate),
FOREIGN KEY (id_dimLocation) REFERENCES dimlocation(id_dimLocation)
);

4. **Optimalisasi dan Standardisasi Charset Database**

Run SQL query/queries on database dw_sales:

```
ALTER DATABASE dw_sales CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
ALTER TABLE dimcustomer CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
ALTER TABLE dimproducts CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
ALTER TABLE dimdate CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
ALTER TABLE dimlocation CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
ALTER TABLE fact_orders CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0012 seconds.)

`ALTER DATABASE dw_sales CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;`

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0078 seconds.)

`ALTER TABLE dimcustomer CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;`

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0119 seconds.)

`ALTER TABLE dimproducts CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;`

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0038 seconds.)

`ALTER TABLE dimdate CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;`

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0092 seconds.)

`ALTER TABLE dimlocation CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;`

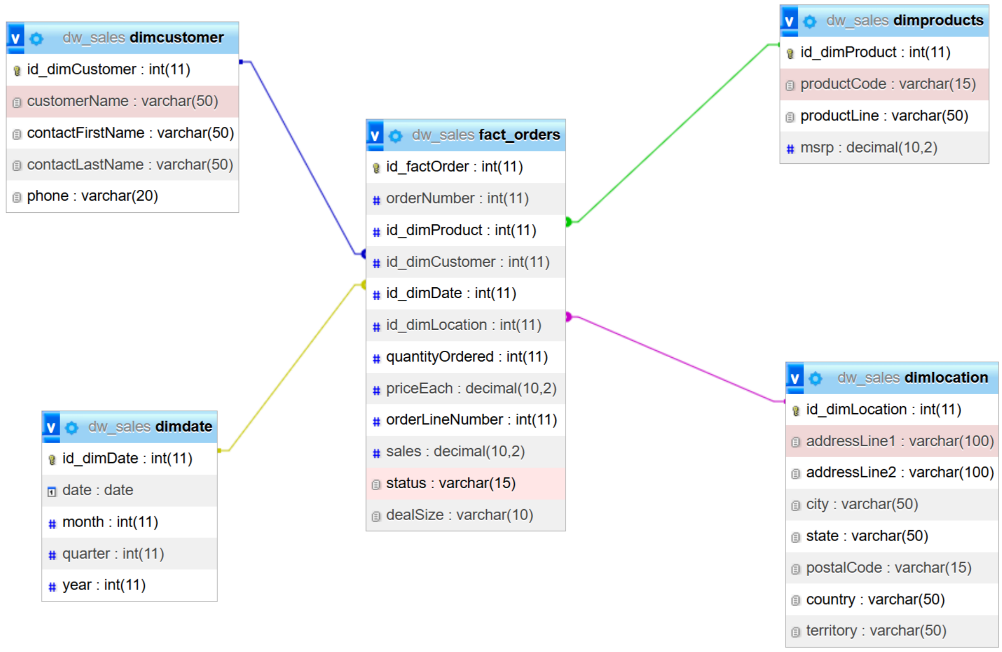
[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

`ALTER TABLE fact_orders CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;`

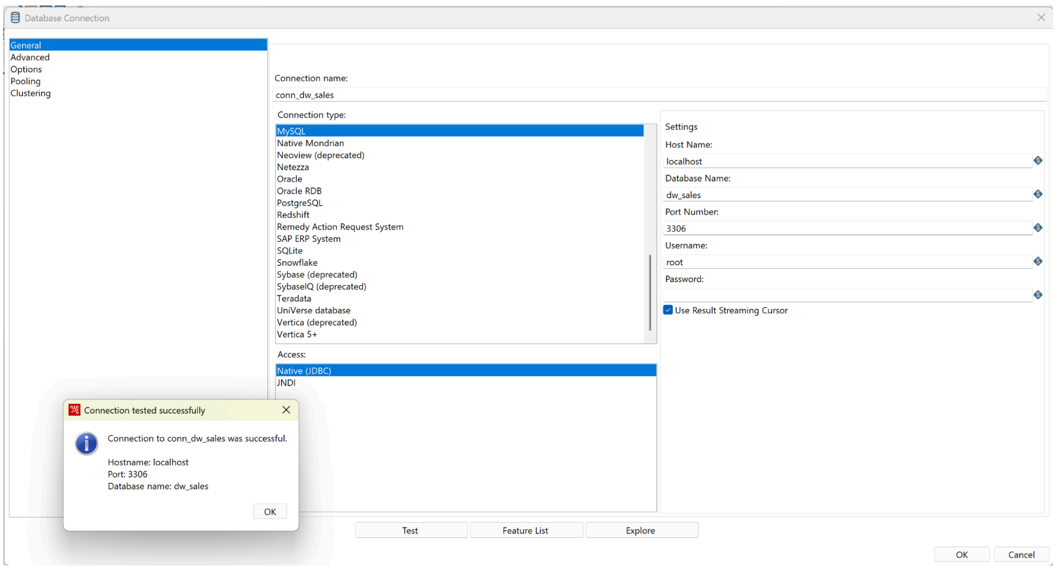
[Edit inline] [Edit] [Create PHP code]

5. Design Database



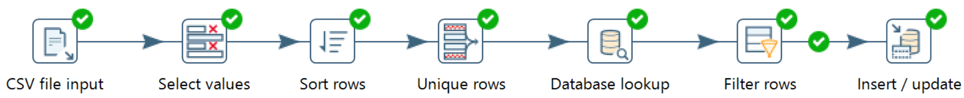
D. Tahap Transform (Transformasi Data)

1. Create connection database

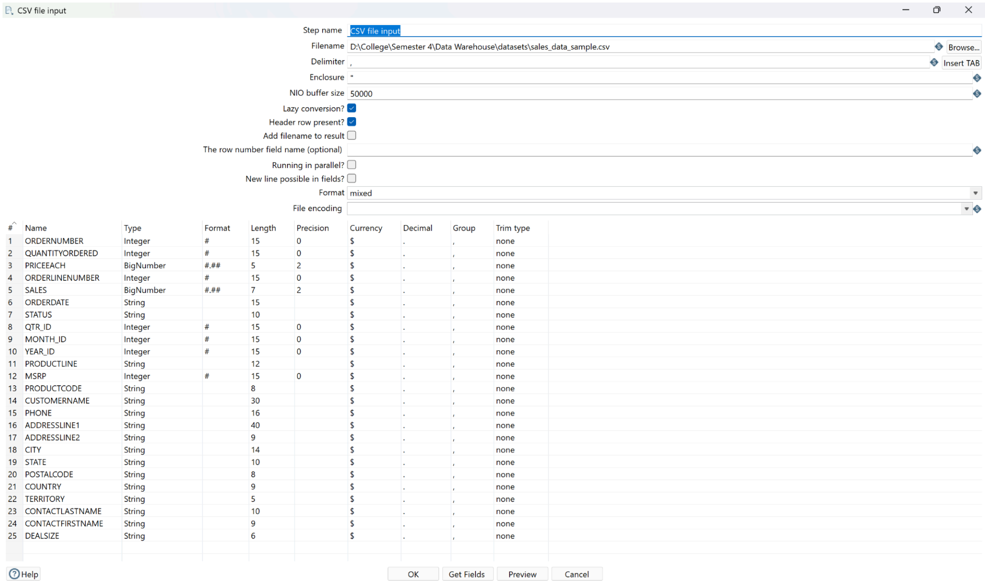


2. Transformasi dim Products

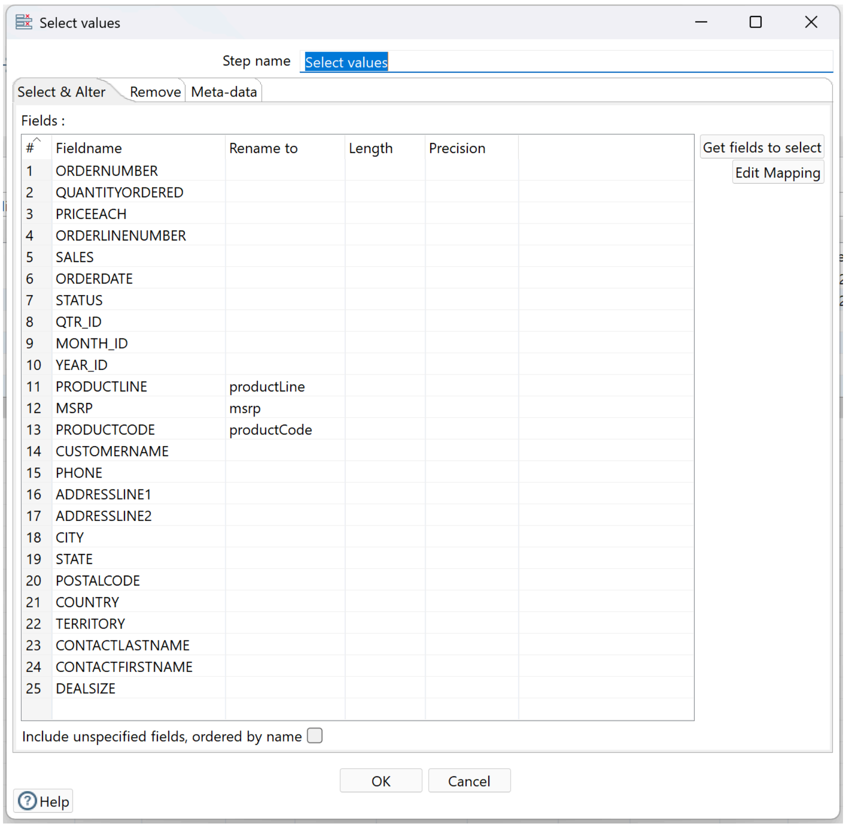
1. Membuat object



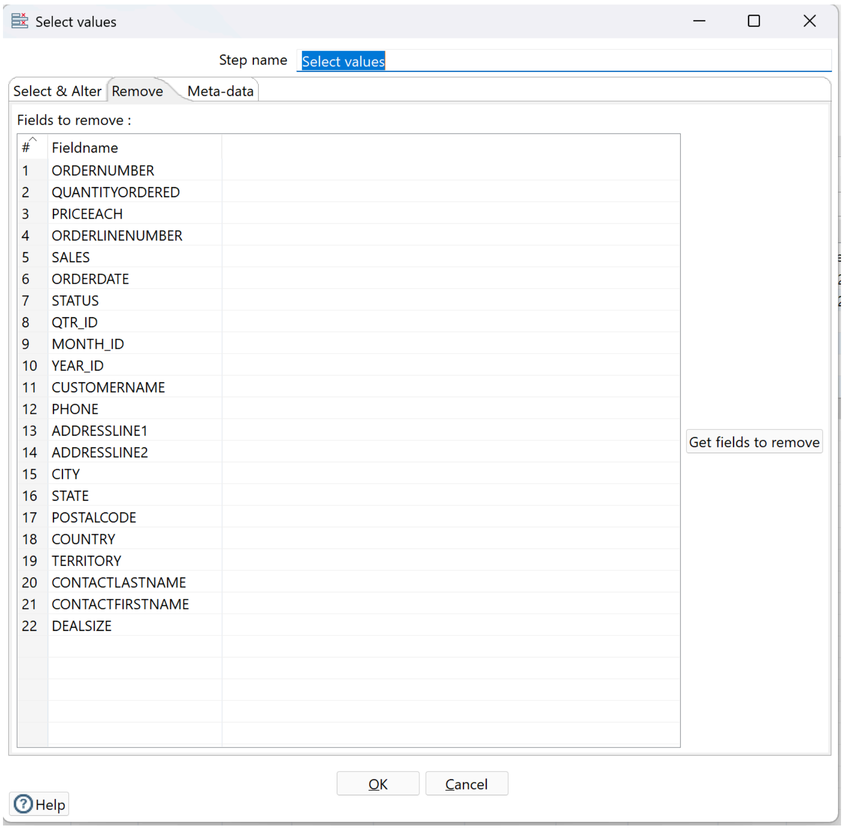
2. Konfigurasi CSV file input



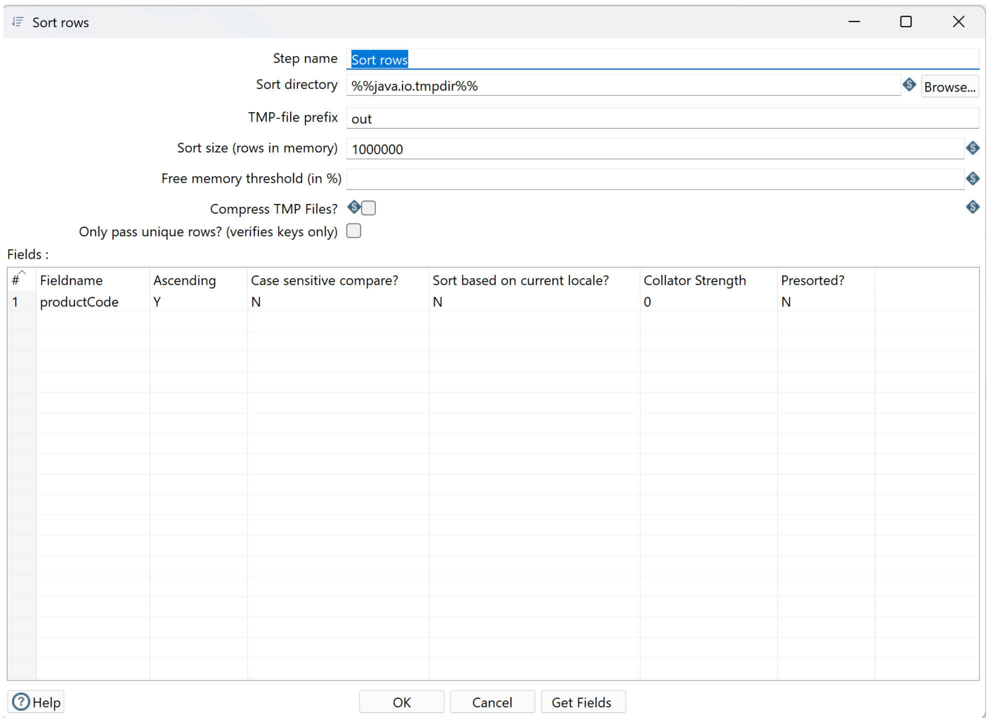
3. Select values
   - Select & alter



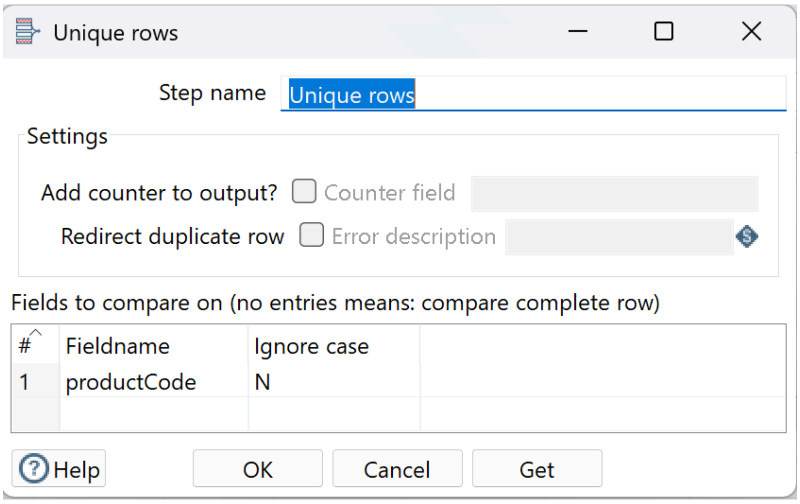
- Remove



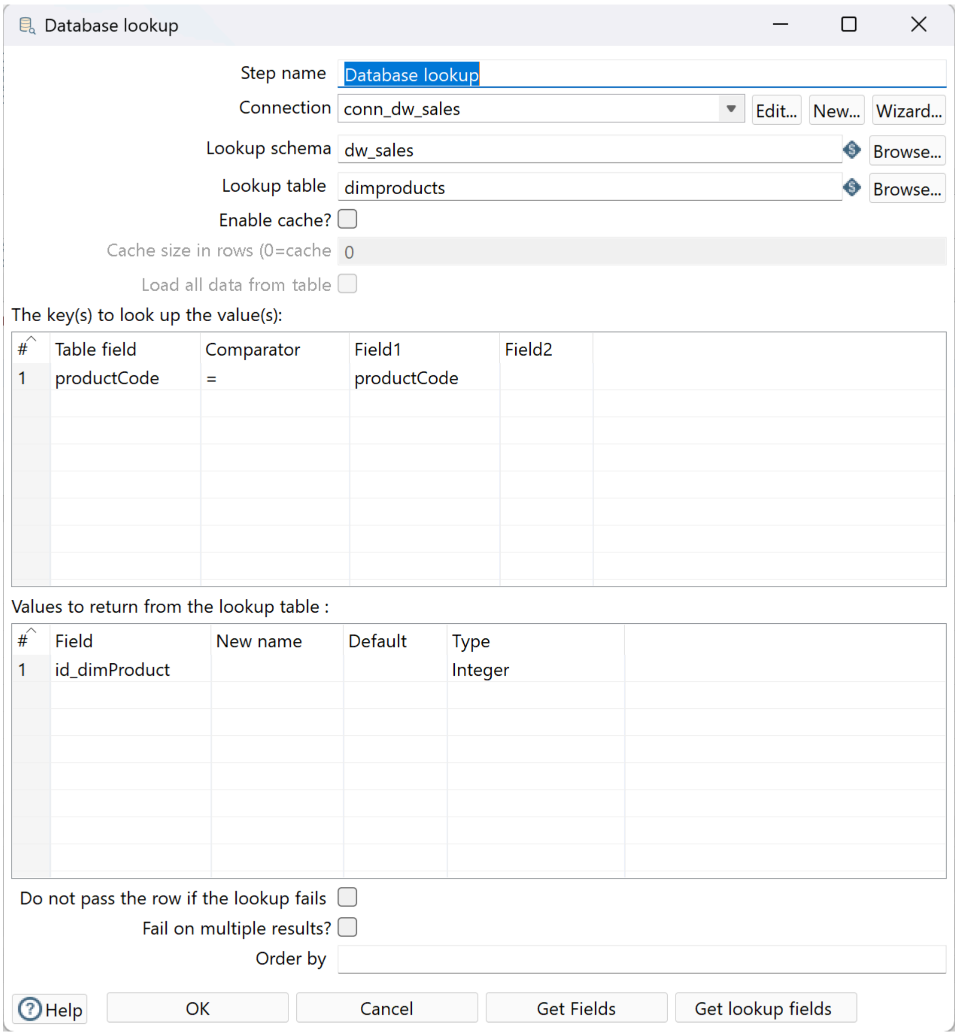
4. Sort rows



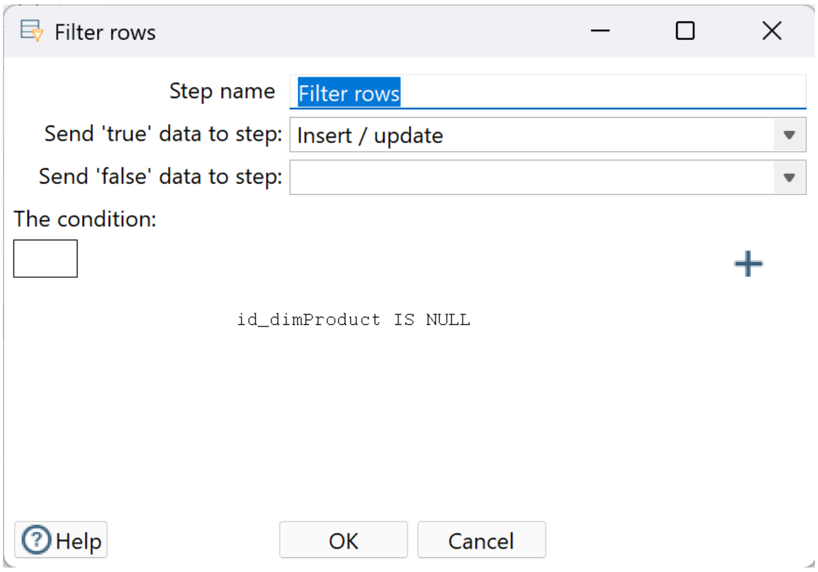
5. Unique rows



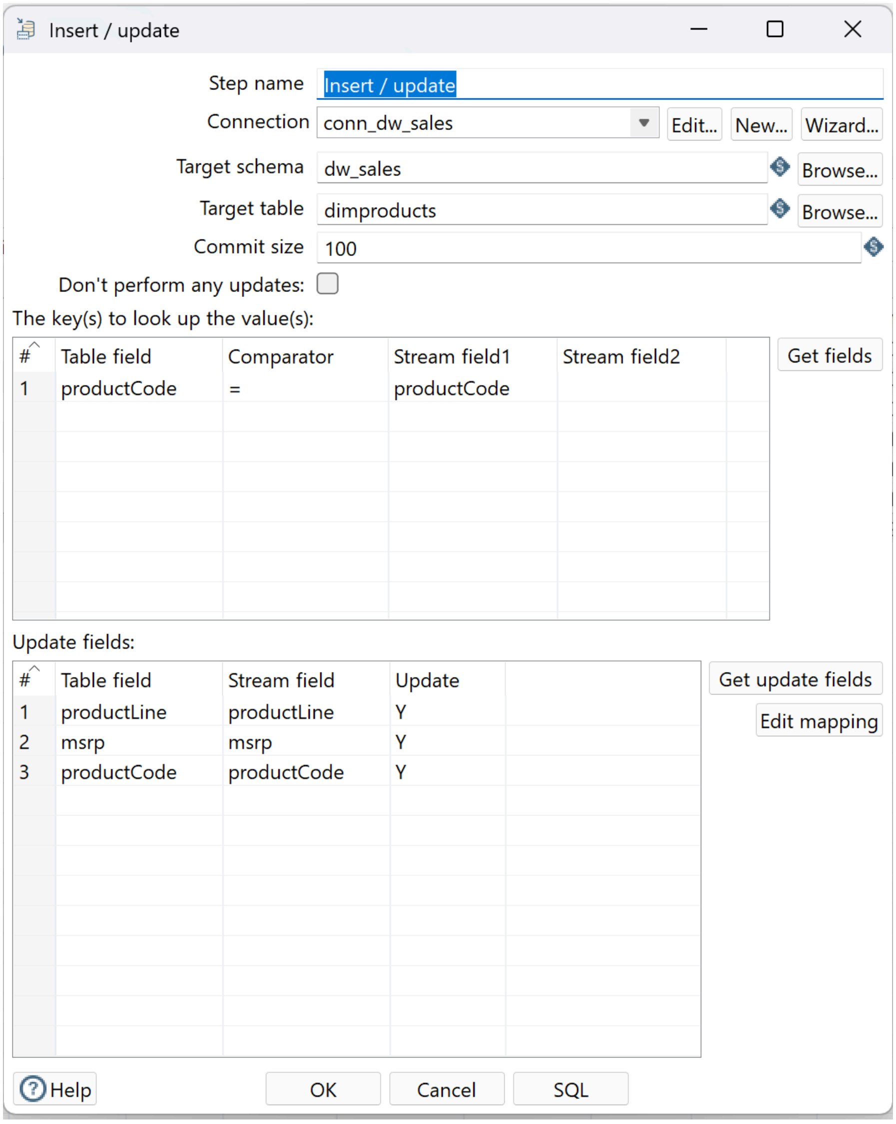
6. Database lookup



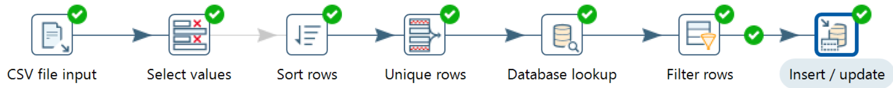
7. Filter rows



8. Insert / Update



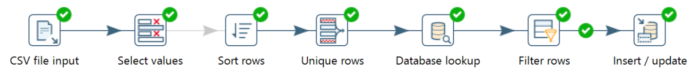
9. Hasil run



Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

2026-06-03 20:41:40.442 - Spoon - Save file as...
2026-06-03 20:41:40.477 - Spoon - Transformation opened.
2026-06-03 20:41:40.477 - Spoon - Launching transformation [dimproducts_project uas]...
2026-06-03 20:41:40.477 - Spoon - Started the transformation execution.
2026-06-03 20:41:40.797 - dimproducts_project uas - Dispatching started for transformation [dimproducts_project uas]
2026-06-03 20:41:40.846 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 20:41:40.853 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 20:41:40.857 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:41:40.878 - Sort rows.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:41:40.896 - Unique rows.0 - Finished processing (I=0, O=0, R=2823, W=109, U=0, E=0)
2026-06-03 20:41:40.937 - Database lookup.0 - Finished processing (I=0, O=0, R=109, W=109, U=0, E=0)
2026-06-03 20:41:40.956 - Filter rows.0 - Finished processing (I=0, O=0, R=109, W=109, U=0, E=0)
2026-06-03 20:41:41.020 - Insert / update.0 - Finished processing (I=109, O=109, R=109, W=109, U=0, E=0)
2026-06-03 20:41:41.021 - Spoon - The transformation has finished!!



Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	282,400	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.0s	201,643	-
3	Sort rows	0	2823	2823	0	0	0	0	0	Finished	0.0s	78,417	-
4	Unique rows	0	2823	109	0	0	0	0	0	Finished	0.1s	52,278	-
5	Database lookup	0	109	109	0	0	0	0	0	Finished	0.1s	1,160	-
6	Filter rows	0	109	109	0	0	0	0	0	Finished	0.1s	965	-
7	Insert / update	0	109	109	109	109	0	0	0	Finished	0.2s	612	-

Showing rows 0 - 24 (109 total, Query took 0.0007 seconds.) [id_dimProduct: **109... - 85...**]

```
SELECT * FROM `dimproducts` ORDER BY `dimproducts`.`id_dimProduct` DESC
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | Show all | Number of rows: 25 | Filter rows: Search this tal

Extra options

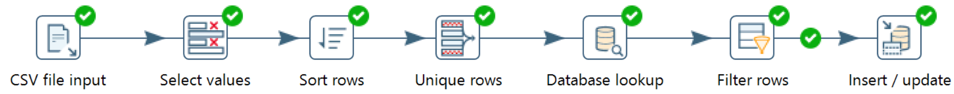
	id_dimProduct	productCode	productLine	msrp
Edit Copy Delete	109	S72_3212	Ships	54.00
Edit Copy Delete	108	S72_1253	Planes	49.00
Edit Copy Delete	107	S700_4002	Planes	74.00
Edit Copy Delete	106	S700_3962	Ships	99.00
Edit Copy Delete	105	S700_3505	Ships	100.00
Edit Copy Delete	104	S700_3167	Planes	80.00
Edit Copy Delete	103	S700_2834	Planes	118.00
Edit Copy Delete	102	S700_2824	Classic Cars	101.00
Edit Copy Delete	101	S700_2610	Ships	72.00
Edit Copy Delete	100	S700_2466	Planes	99.00
Edit Copy Delete	99	S700_2047	Ships	90.00
Edit Copy Delete	98	S700_1938	Ships	86.00
Edit Copy Delete	97	S700_1691	Planes	91.00
Edit Copy Delete	96	S700_1138	Ships	66.00
Edit Copy Delete	95	S50_4713	Motorcycles	81.00
Edit Copy Delete	94	S50_1514	Trains	58.00
Edit Copy Delete	93	S50_1392	Trucks and Buses	115.00
Edit Copy Delete	92	S50_1341	Vintage Cars	43.00
Edit Copy Delete	91	S32_4485	Motorcycles	102.00
Edit Copy Delete	90	S32_4289	Vintage Cars	68.00

Console

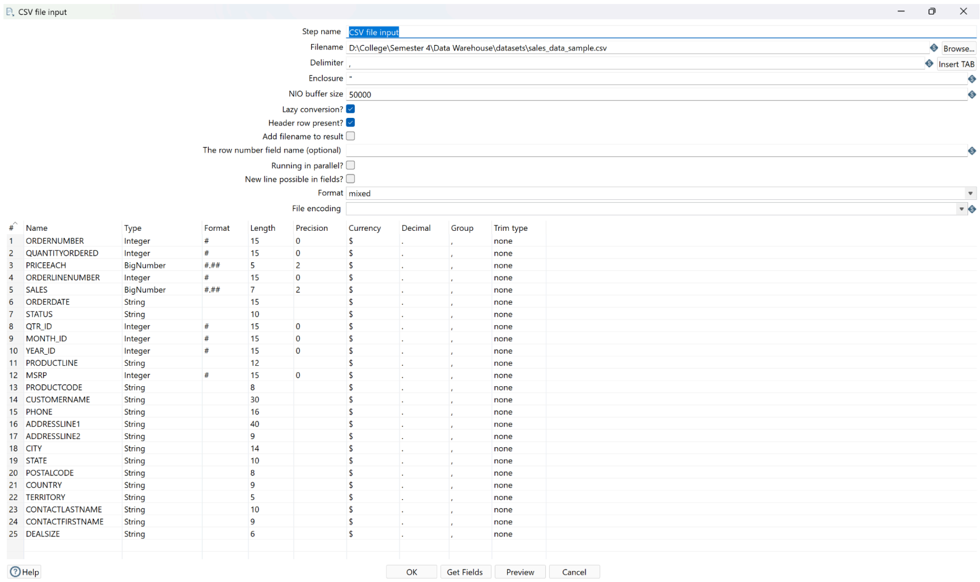
10. Penjelasan Hasil : Hasil ETL menunjukkan bahwa proses transformasi dimProducts berhasil dijalankan tanpa error. Dari 2.824 baris data yang dibaca dari file CSV, setelah melalui proses deduplikasi berdasarkan productCode diperoleh 109 baris unik yang kemudian berhasil di-insert ke tabel dimproducts. Hal ini membuktikan bahwa proses select values, sort rows, unique rows, database lookup, dan insert/update berjalan dengan baik sesuai pipeline yang dirancang.

3. Transformasi dim Customer

1. Membuat object



2. Konfigurasi CSV file input



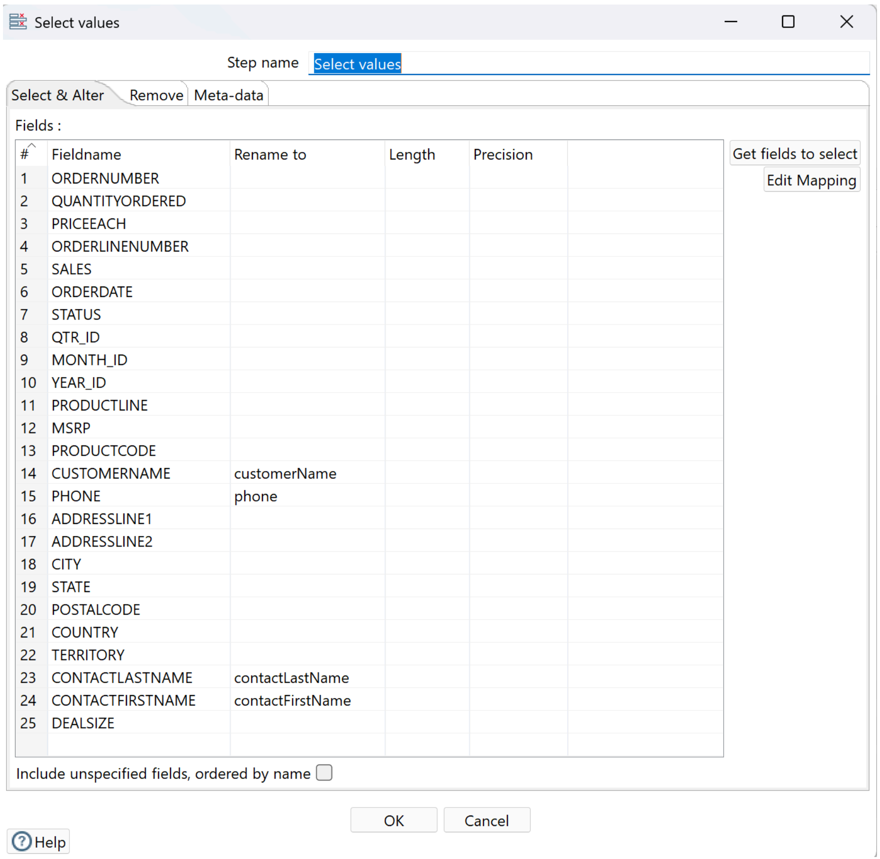
CSV file input

Step name: CSV file input
Filename: D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv — Browse...
Delimiter: , — Insert TAB
Enclosure: "
NIO buffer size: 50000
Lazy conversion? ☑
Header row present? ☑
Add filename to result ☐
The row number field name (optional):
Running in parallel? ☐
New line possible in fields? ☐
Format: mixed
File encoding:

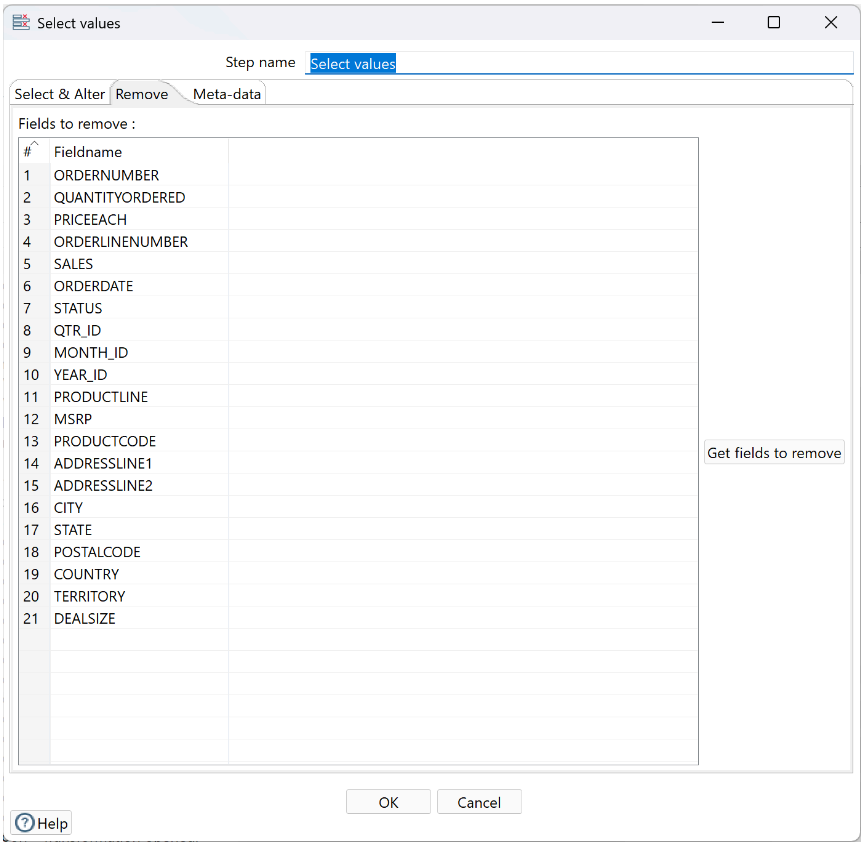
#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	$	.	,	none
2	QUANTITYORDERED	Integer	#	15	0	$	.	,	none
3	PRICEEACH	BigNumber	#.##	5	2	$	.	,	none
4	ORDERLINENUMBER	Integer	#	15	0	$	.	,	none
5	SALES	BigNumber	#.##	7	2	$	.	,	none
6	ORDERDATE	String		15		$	.	,	none
7	STATUS	String		10		$	.	,	none
8	QTR_ID	Integer	#	15	0	$	.	,	none
9	MONTH_ID	Integer	#	15	0	$	.	,	none
10	YEAR_ID	Integer	#	15	0	$	.	,	none
11	PRODUCTLINE	String		12		$	.	,	none
12	MSRP	Integer	#	15	0	$	.	,	none
13	PRODUCTCODE	String		8		$	.	,	none
14	CUSTOMERNAME	String		30		$	.	,	none
15	PHONE	String		16		$	.	,	none
16	ADDRESSLINE1	String		40		$	.	,	none
17	ADDRESSLINE2	String		9		$	.	,	none
18	CITY	String		14		$	.	,	none
19	STATE	String		10		$	.	,	none
20	POSTALCODE	String		8		$	.	,	none
21	COUNTRY	String		9		$	.	,	none
22	TERRITORY	String		5		$	.	,	none
23	CONTACTLASTNAME	String		10		$	.	,	none
24	CONTACTFIRSTNAME	String		9		$	.	,	none
25	DEALSIZE	String		6		$	.	,	none

Help — OK — Get Fields — Preview — Cancel

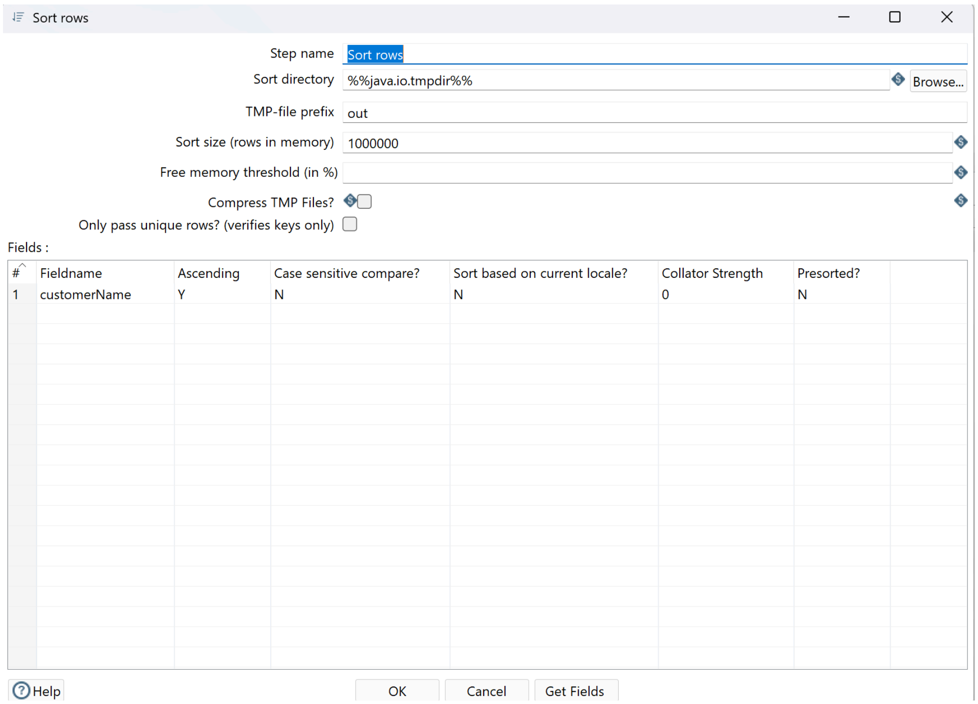
3. Select values
   - Select & alter



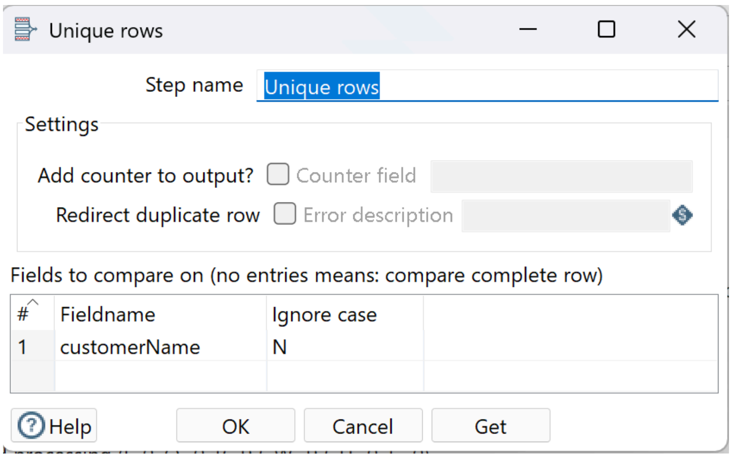
- Remove



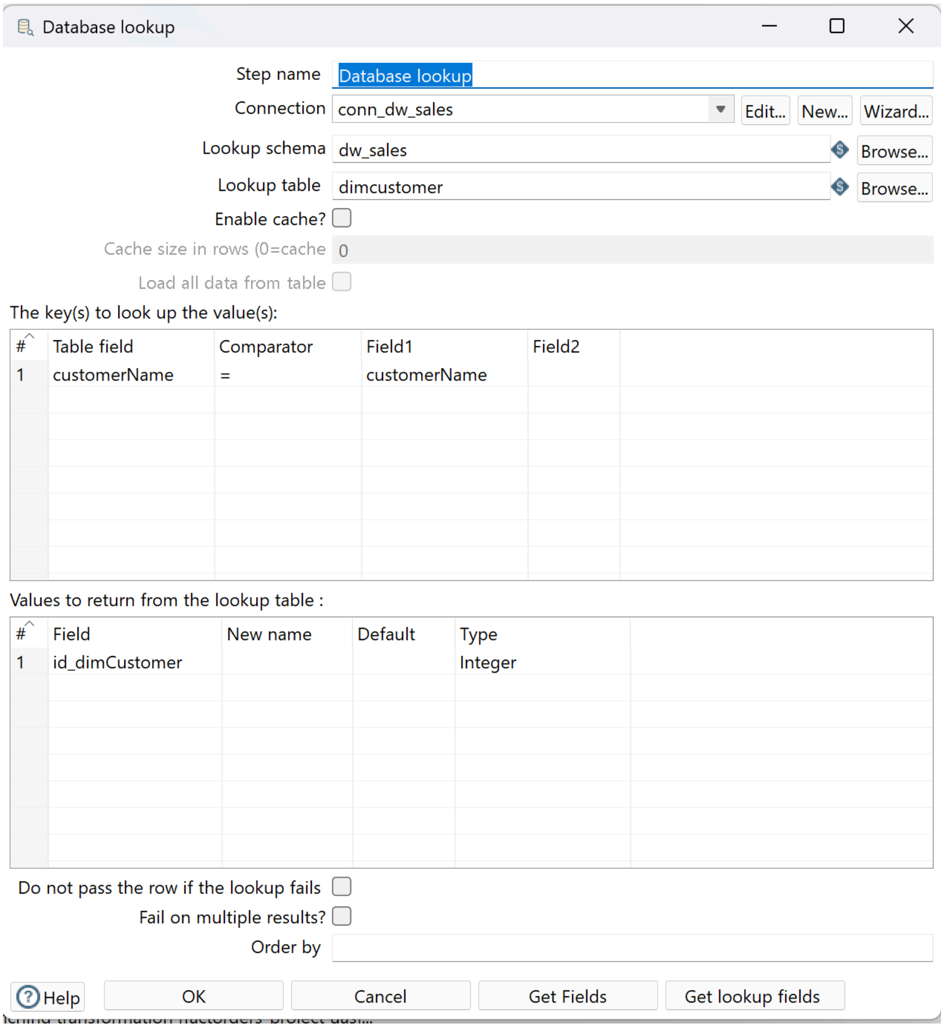
4. Sort rows



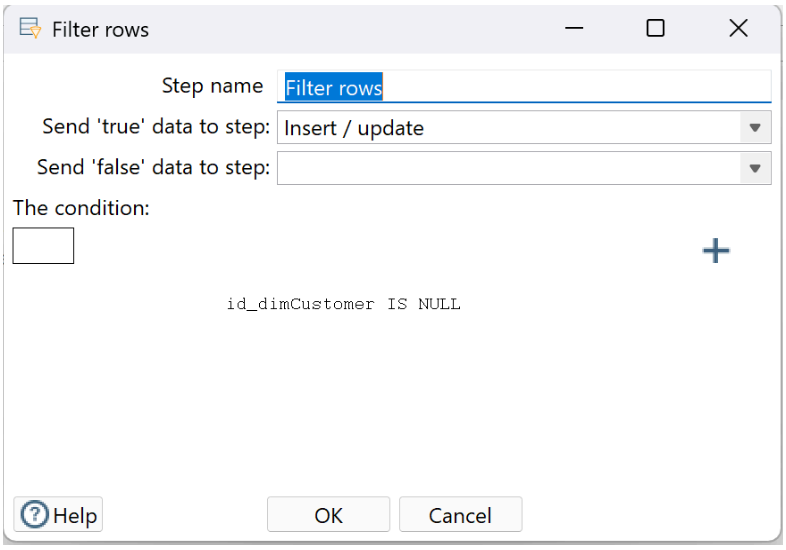
5. Unique rows



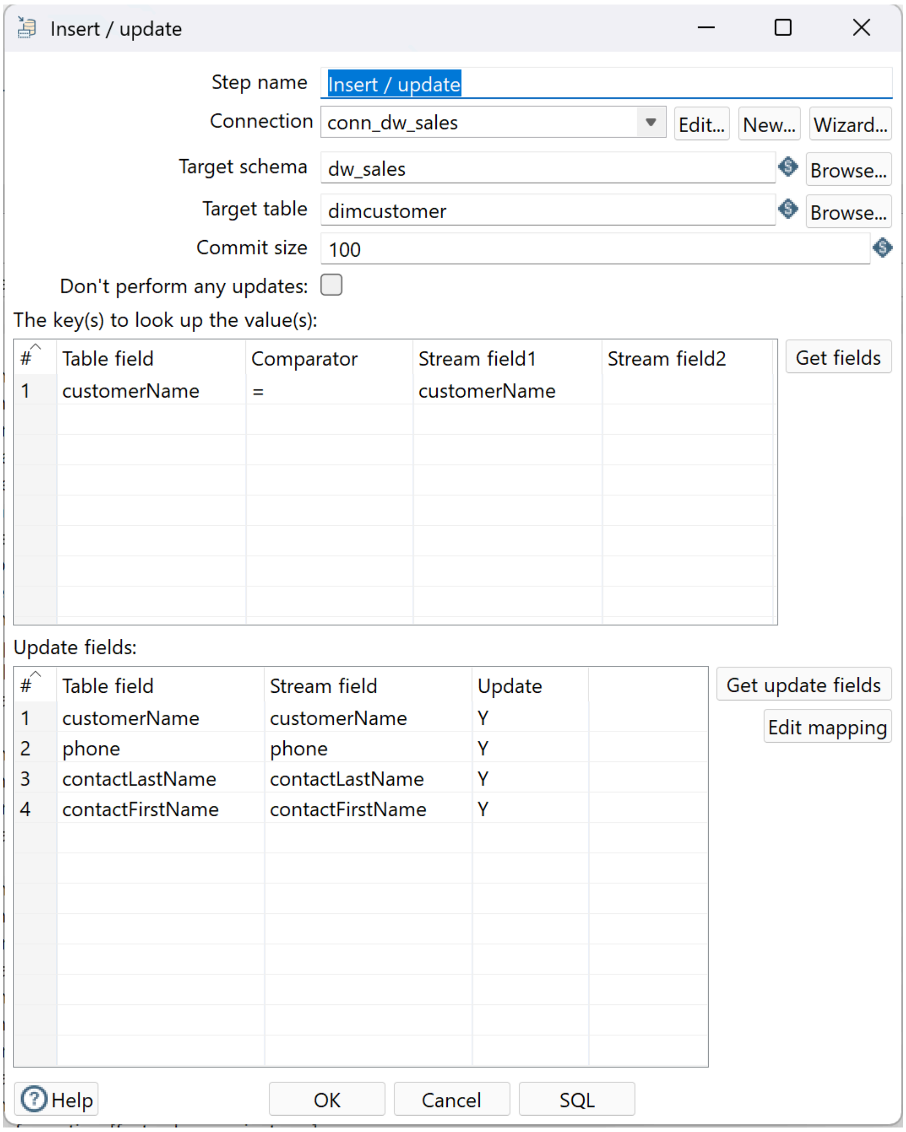
6. Database lookup



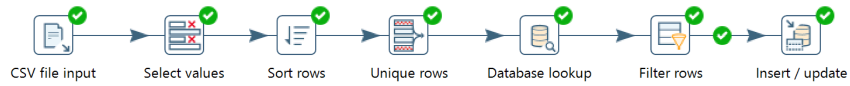
7. Filter rows



8. Insert / Update



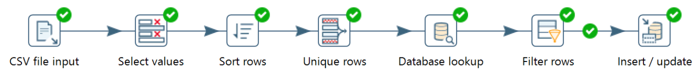
9. Hasil run



Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

```
2026-06-03 20:40:54.451 - Spoon - Save file as...
2026-06-03 20:40:54.504 - Spoon - Transformation opened.
2026-06-03 20:40:54.504 - Spoon - Launching transformation [dimcustomer_project uas]...
2026-06-03 20:40:54.504 - Spoon - Started the transformation execution.
2026-06-03 20:40:54.835 - dimcustomer_project uas - Dispatching started for transformation [dimcustomer_project uas]
2026-06-03 20:40:54.886 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 20:40:54.895 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 20:40:54.898 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:40:54.933 - Sort rows.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:40:54.947 - Unique rows.0 - Finished processing (I=0, O=0, R=2823, W=92, U=0, E=0)
2026-06-03 20:40:54.996 - Database lookup.0 - Finished processing (I=0, O=0, R=92, W=92, U=0, E=0)
2026-06-03 20:40:55.009 - Filter rows.0 - Finished processing (I=0, O=0, R=92, W=92, U=0, E=0)
2026-06-03 20:40:55.073 - Insert / update.0 - Finished processing (I=92, O=92, R=92, W=92, U=0, E=0)
2026-06-03 20:40:55.074 - Spoon - The transformation has finished!!
```



Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	235,333	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.0s	176,437	-
3	Sort rows	0	2823	2823	0	0	0	0	0	Finished	0.1s	55,353	-
4	Unique rows	0	2823	92	0	0	0	0	0	Finished	0.1s	43,431	-
5	Database lookup	0	92	92	0	0	0	0	0	Finished	0.1s	814	-
6	Filter rows	0	92	92	0	0	0	0	0	Finished	0.1s	730	-
7	Insert / update	0	92	92	92	92	0	0	0	Finished	0.2s	482	-

✔ Showing rows 0 - 24 (92 total, Query took 0.0004 seconds.) [id_dimCustomer: **92... - 68...**]

```
SELECT * FROM `dimcustomer` ORDER BY `dimcustomer`.`id_dimCustomer` DESC
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

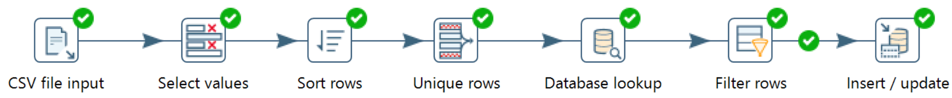
	id_dimCustomer	customerName	contactFirstName	contactLastName	phone
Edit Copy Delete	92	West Coast Collectables Co.	Steve	Thompson	3105553722
Edit Copy Delete	91	Volvo Model Replicas, Co	Christina	Berglund	0921-12 3555
Edit Copy Delete	90	Vitachrome Inc.	Michael	Frick	2125551500
Edit Copy Delete	89	Vida Sport, Ltd	Michael	Holz	0897-034555
Edit Copy Delete	88	UK Collectables, Ltd.	Elizabeth	Devon	(171) 555-2282
Edit Copy Delete	87	Toys4GrownUps.com	Julie	Young	6265557265
Edit Copy Delete	86	Toys of Finland, Co.	Matti	Karttunen	90-224 8555
Edit Copy Delete	85	Toms Spezialitten, Ltd	Henriette	Pfalzheim	0221-5554327
Edit Copy Delete	84	Tokyo Collectables, Ltd	Akiko	Shimamura	+81 3 3584 0555
Edit Copy Delete	83	The Sharp Gifts Warehouse	Sue	Frick	4085553659
Edit Copy Delete	82	Tekni Collectables Inc.	William	Brown	2015559350
Edit Copy Delete	81	Technics Stores Inc.	Juri	Hirano	6505556809
Edit Copy Delete	80	Super Scale Inc.	Leslie	Murphy	2035559545
Edit Copy Delete	79	Suominen Souveniers	Kalle	Suominen	+358 9 8045 555
Edit Copy Delete	78	Stylish Desk Decors, Co.	Ann	Brown	(171) 555-0297
Edit Copy Delete	77	Souveniers And Things Co.	Adrian	Huxley	+61 2 9495 8555
Edit Copy Delete	76	Signal Gift Stores	Sue	King	7025551838
Edit Copy Delete	75	Signal Collectibles Ltd.	Sue	Taylor	4155554312
Edit Copy Delete	74	Scandinavian Gift Ideas	Maria	Larsson	0695-34 6555
Edit Copy Delete	73	Saveley & Henriot, Co.	Mary	Saveley	78.32.5555

Console

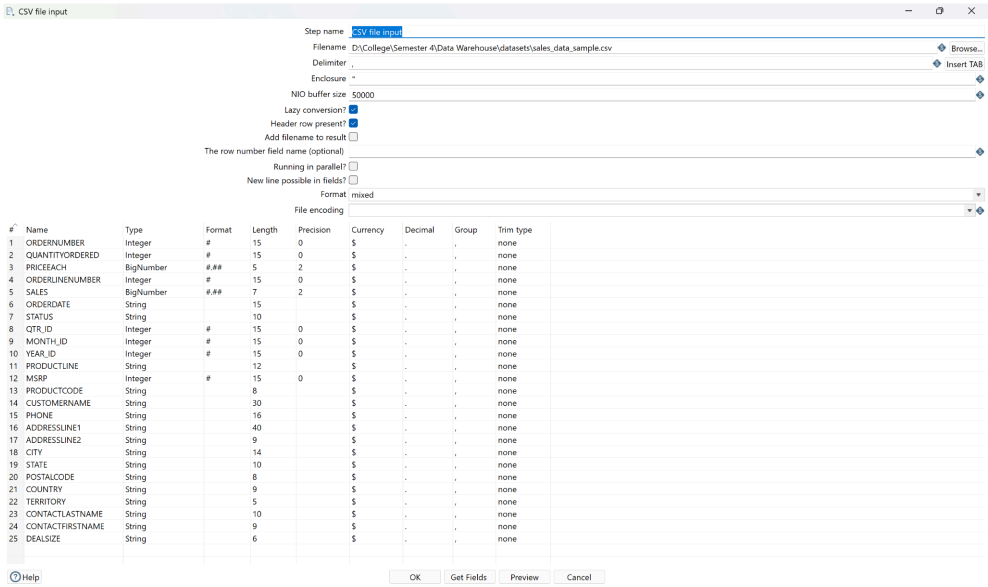
10. Penjelasan Hasil : Hasil ETL menunjukkan bahwa proses transformasi dimCustomer berhasil dijalankan tanpa error. Dari 2.823 baris data valid, setelah deduplikasi berdasarkan customerName diperoleh 92 baris unik yang berhasil di-insert ke tabel dimcustomer. Hal ini membuktikan bahwa setiap pelanggan hanya tersimpan satu kali dalam tabel dimensi sesuai dengan desain star schema yang telah dirancang.

4. Transformasi dim Date

1. Membuat object



2. Konfigurasi CSV file input



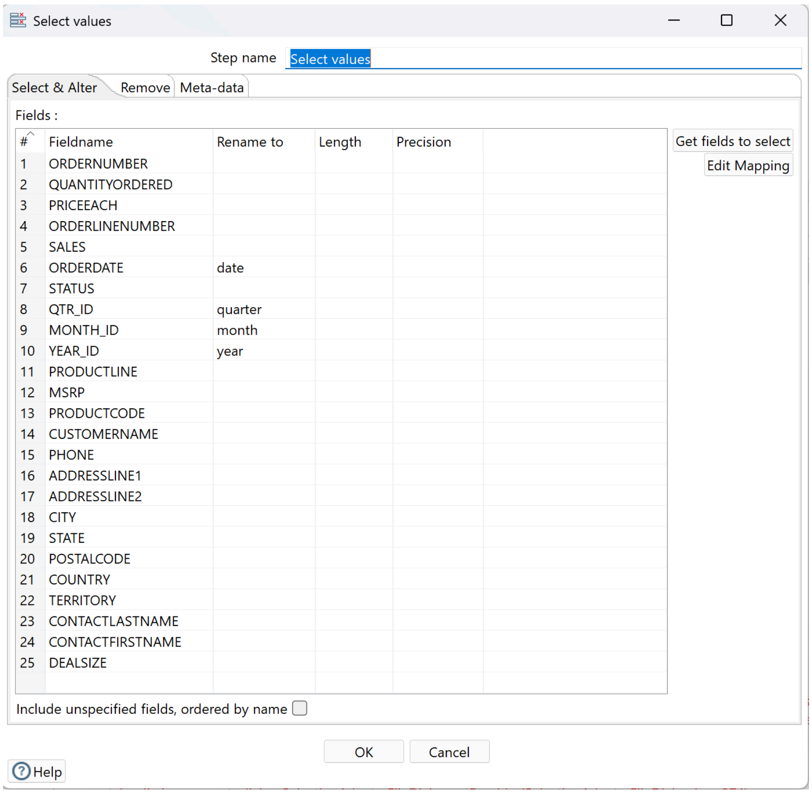
CSV file input

Step name: CSV file input
Filename: D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv
Delimiter: ,
Enclosure: "
NIO buffer size: 50000
Lazy conversion? ☑
Header row present? ☑
Add filename to result ☐
The row number field name (optional)
Running in parallel? ☐
New line possible in fields? ☐
Format: mixed
File encoding

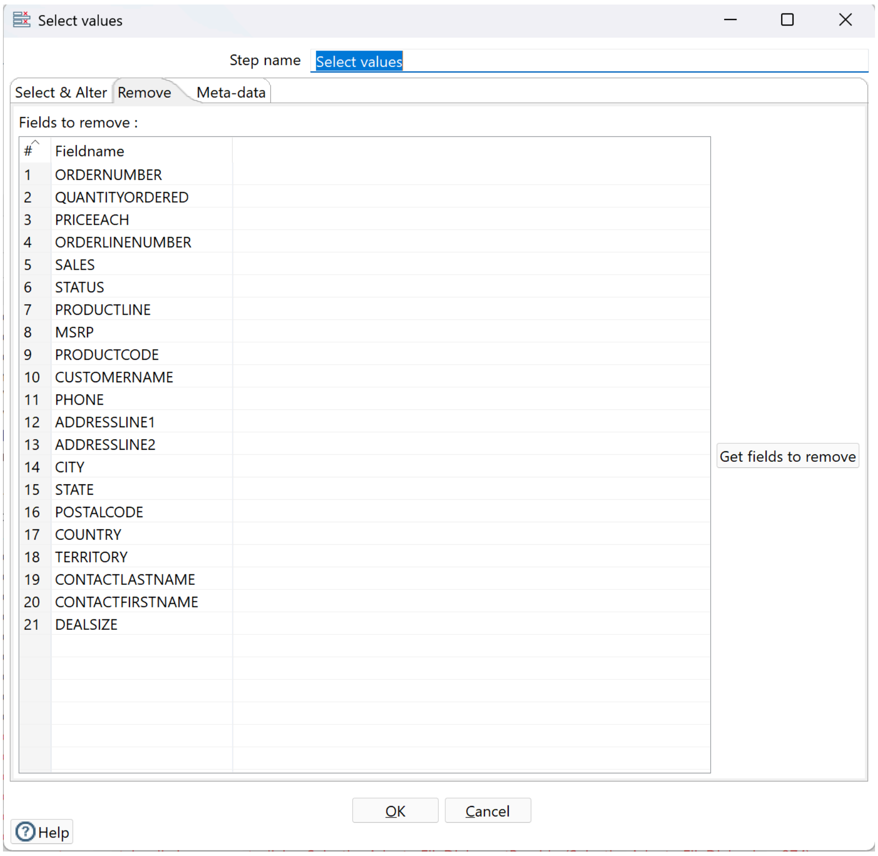
#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	$	.	,	none
2	QUANTITYORDERED	Integer	#	15	0	$	.	,	none
3	PRICEEACH	BigNumber	#.##	5	2	$	.	,	none
4	ORDERLINENUMBER	Integer	#	15	0	$	.	,	none
5	SALES	BigNumber	#.##	7	2	$	.	,	none
6	ORDERDATE	String		15		$	.	,	none
7	STATUS	String		10		$	.	,	none
8	QTR_ID	Integer	#	15	0	$	.	,	none
9	MONTH_ID	Integer	#	15	0	$	.	,	none
10	YEAR_ID	Integer	#	15	0	$	.	,	none
11	PRODUCTLINE	String		12		$	.	,	none
12	MSRP	Integer	#	15	0	$	.	,	none
13	PRODUCTCODE	String		8		$	.	,	none
14	CUSTOMERNAME	String		30		$	.	,	none
15	PHONE	String		16		$	.	,	none
16	ADDRESSLINE1	String		40		$	.	,	none
17	ADDRESSLINE2	String		9		$	.	,	none
18	CITY	String		14		$	.	,	none
19	STATE	String		10		$	.	,	none
20	POSTALCODE	String		8		$	.	,	none
21	COUNTRY	String		9		$	.	,	none
22	TERRITORY	String		5		$	.	,	none
23	CONTACTLASTNAME	String		10		$	.	,	none
24	CONTACTFIRSTNAME	String		9		$	.	,	none
25	DEALSIZE	String		6		$	.	,	none

Help | OK | Get Fields | Preview | Cancel

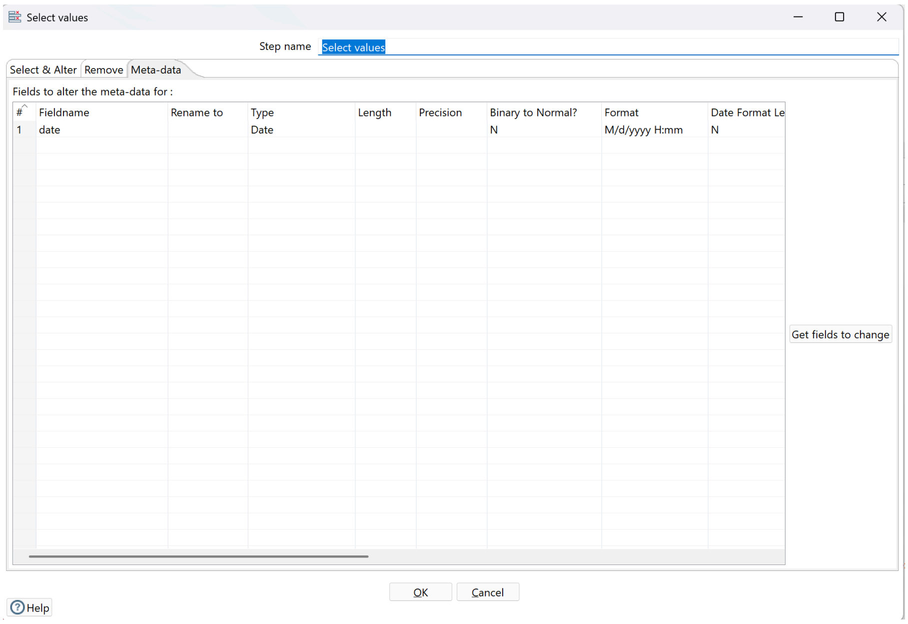
3. Select values
   - Select & alter



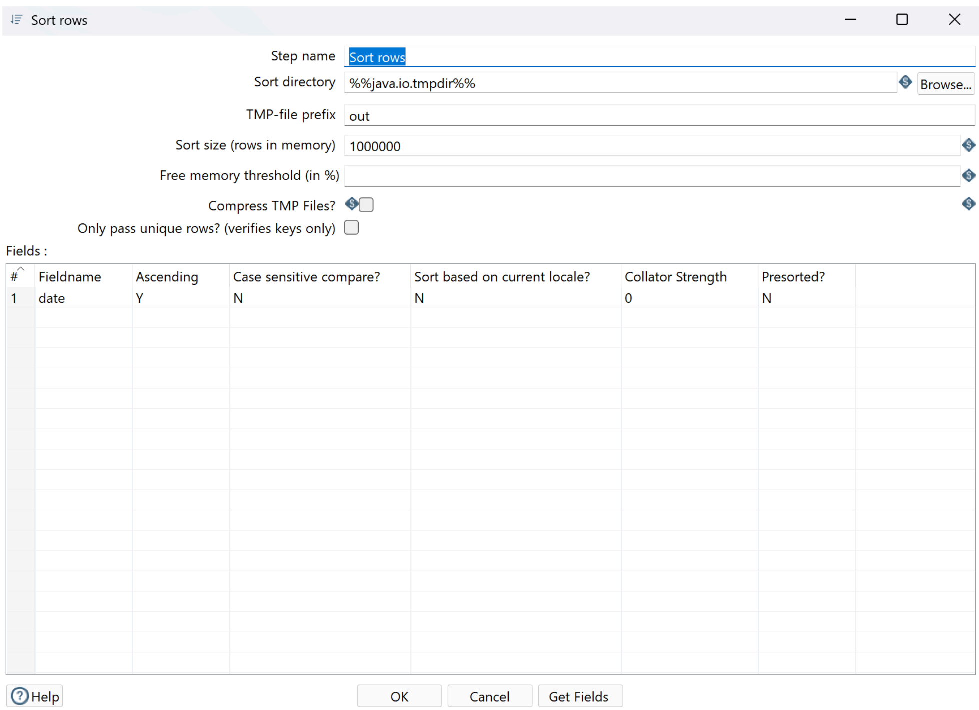
- Remove



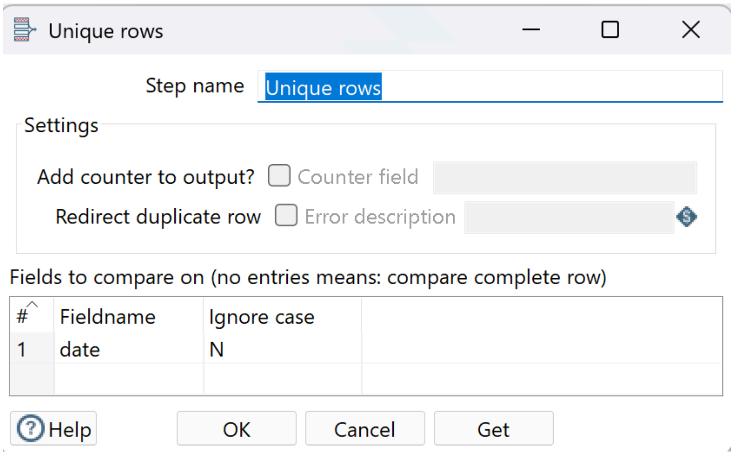
- Meta data



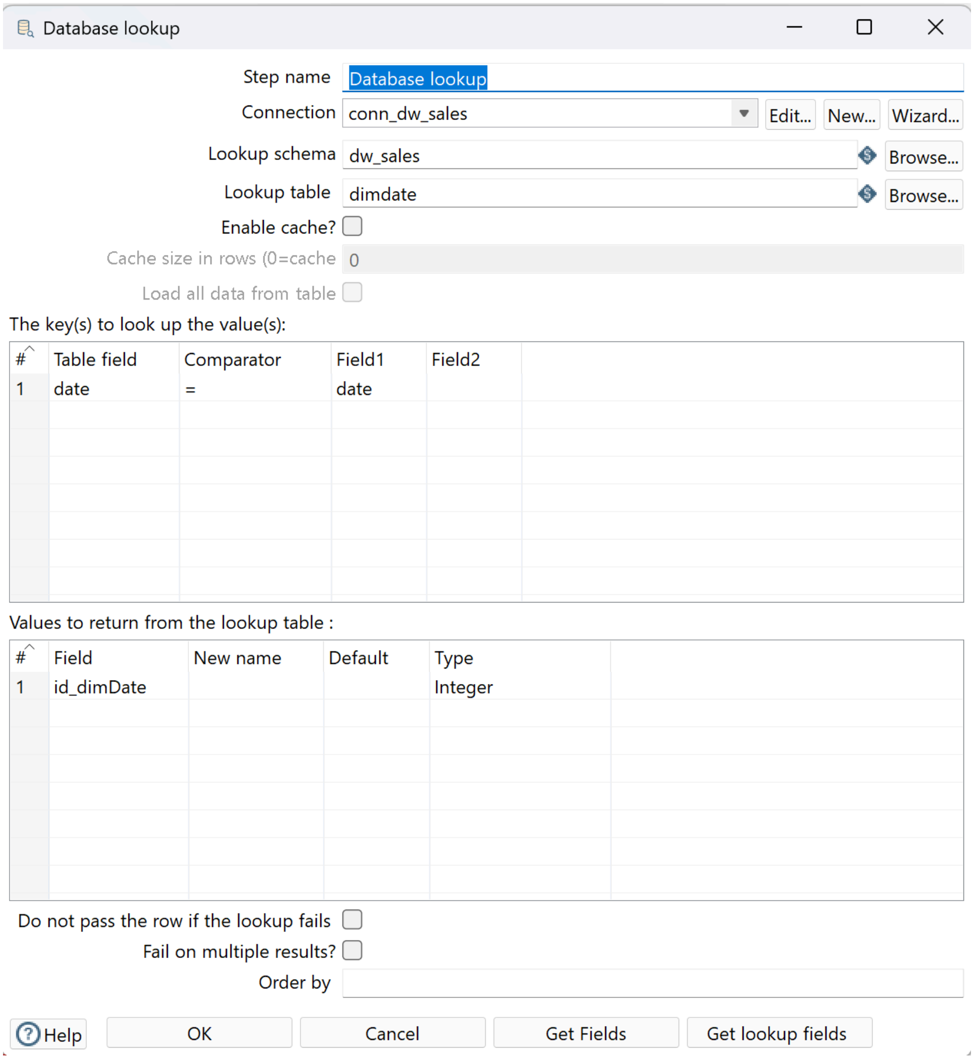
4. Sort rows



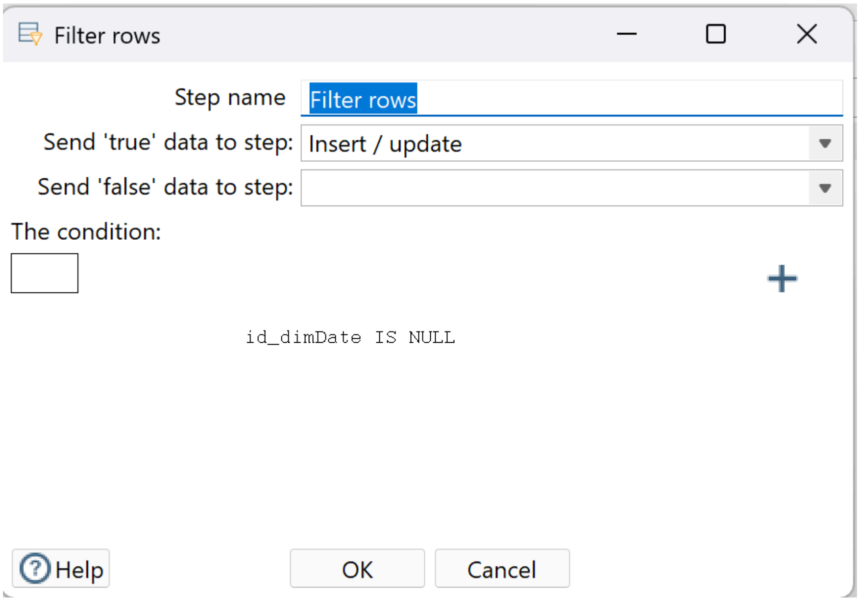
5. Unique rows



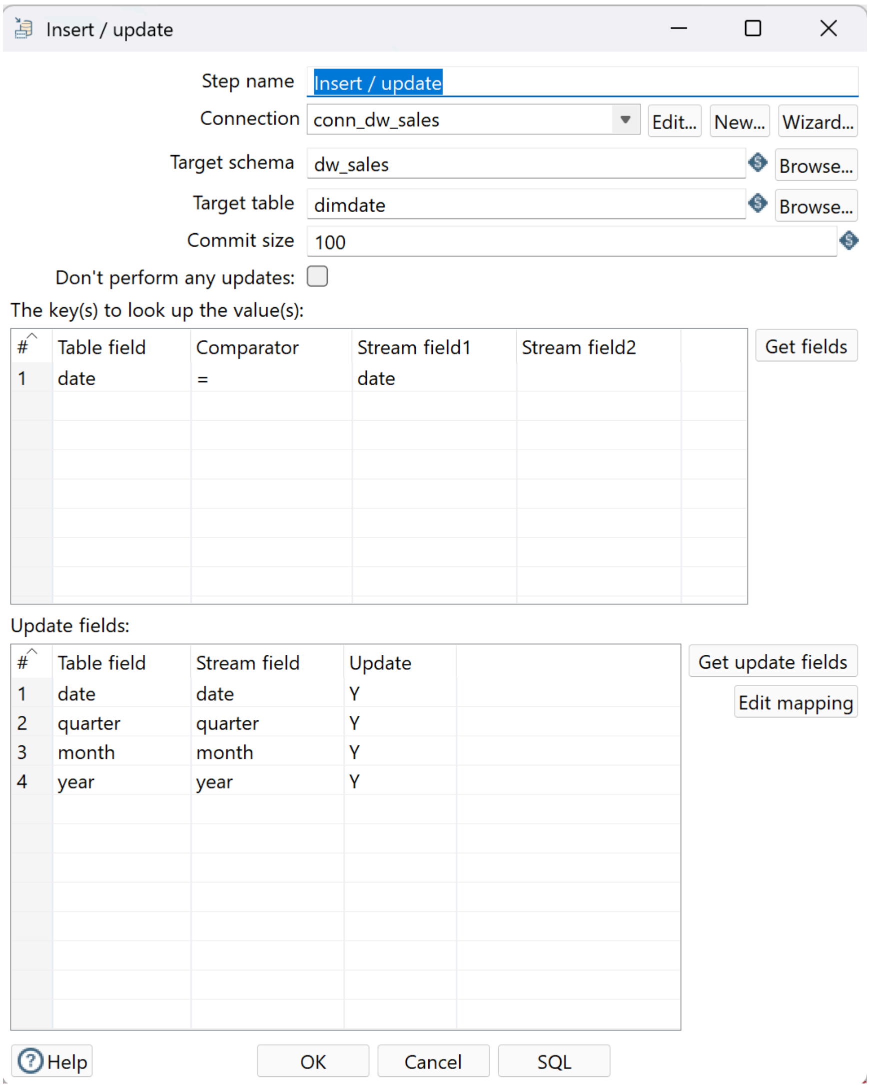
6. Database lookup



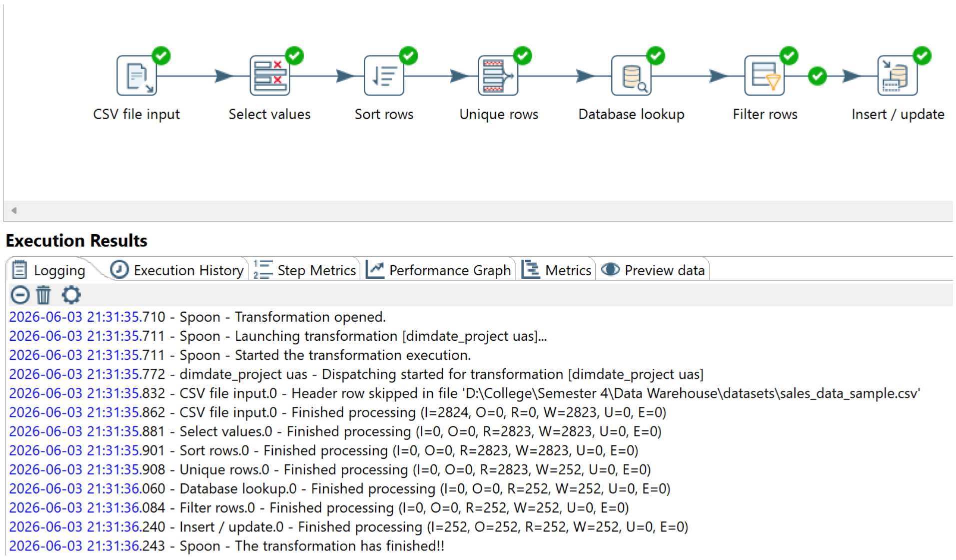
7. Filter rows

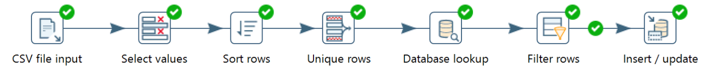


8. Insert / Update


Insert / update
Step name
Insert / update
Connection
conn_dw_sales
Edit...
New...
Wizard...
Target schema
dw_sales
Browse...
Target table
dimdate
Browse...
Commit size
100
Don't perform any updates:
The key(s) to look up the value(s):
Table field Comparator Stream field1 Stream field2
1 date = date
Get fields
Update fields:
Table field Stream field Update
1 date date Y
2 quarter quarter Y
3 month month Y
4 year year Y
Get update fields
Edit mapping
Help
OK
Cancel
SQL

9. Hasil run


CSV file input
Select values
Sort rows
Unique rows
Database lookup
Filter rows
Insert / update
Execution Results
Logging
Execution History
Step Metrics
Performance Graph
Metrics
Preview data
2026-06-03 21:31:35.710 - Spoon - Transformation opened.
2026-06-03 21:31:35.711 - Spoon - Launching transformation [dimdate_project uas]...
2026-06-03 21:31:35.711 - Spoon - Started the transformation execution.
2026-06-03 21:31:35.772 - dimdate_project uas - Dispatching started for transformation [dimdate_project uas]
2026-06-03 21:31:35.832 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 21:31:35.862 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 21:31:35.881 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 21:31:35.901 - Sort rows.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 21:31:35.908 - Unique rows.0 - Finished processing (I=0, O=0, R=2823, W=252, U=0, E=0)
2026-06-03 21:31:36.060 - Database lookup.0 - Finished processing (I=0, O=0, R=252, W=252, U=0, E=0)
2026-06-03 21:31:36.084 - Filter rows.0 - Finished processing (I=0, O=0, R=252, W=252, U=0, E=0)
2026-06-03 21:31:36.240 - Insert / update.0 - Finished processing (I=252, O=252, R=252, W=252, U=0, E=0)
2026-06-03 21:31:36.243 - Spoon - The transformation has finished!!



Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	76,324	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.1s	50,411	-
3	Sort rows	0	2823	2823	0	0	0	0	0	Finished	0.1s	37,145	-
4	Unique rows	0	2823	252	0	0	0	0	0	Finished	0.1s	34,012	-
5	Database lookup	0	252	252	0	0	0	0	0	Finished	0.2s	1,072	-
6	Filter rows	0	252	252	0	0	0	0	0	Finished	0.3s	969	-
7	Insert / update	0	252	252	252	252	0	0	0	Finished	0.4s	606	-

✔ Showing rows 0 - 24 (252 total, Query took 0.0006 seconds.) [id_dimDate: **252... - 228...**]

```
SELECT * FROM `dimdate` ORDER BY `dimdate`.`id_dimDate` DESC
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | ☐ Show all | Number of rows: 25 Filter rows: Searc

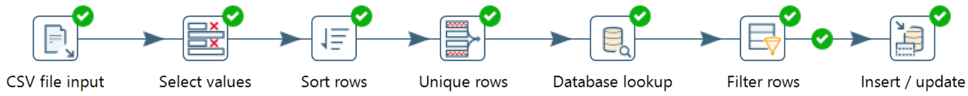
Extra options

←T→				id_dimDate 1	date	month	quarter	year
☐	Edit	Copy	Delete	252	2005-05-31	5	2	2005
☐	Edit	Copy	Delete	251	2005-05-30	5	2	2005
☐	Edit	Copy	Delete	250	2005-05-29	5	2	2005
☐	Edit	Copy	Delete	249	2005-05-17	5	2	2005
☐	Edit	Copy	Delete	248	2005-05-13	5	2	2005
☐	Edit	Copy	Delete	247	2005-05-10	5	2	2005
☐	Edit	Copy	Delete	246	2005-05-09	5	2	2005
☐	Edit	Copy	Delete	245	2005-05-06	5	2	2005
☐	Edit	Copy	Delete	244	2005-05-05	5	2	2005
☐	Edit	Copy	Delete	243	2005-05-03	5	2	2005
☐	Edit	Copy	Delete	242	2005-05-01	5	2	2005
☐	Edit	Copy	Delete	241	2005-04-23	4	2	2005
☐	Edit	Copy	Delete	240	2005-04-22	4	2	2005
☐	Edit	Copy	Delete	239	2005-04-15	4	2	2005
☐	Edit	Copy	Delete	238	2005-04-14	4	2	2005
☐	Edit	Copy	Delete	237	2005-04-08	4	2	2005
☐	Edit	Copy	Delete	236	2005-04-07	4	2	2005
☐	Edit	Copy	Delete	235	2005-04-03	4	2	2005
☐	Edit	Copy	Delete	234	2005-04-01	4	2	2005
Console	it	Copy	Delete	233	2005-03-30	3	1	2005

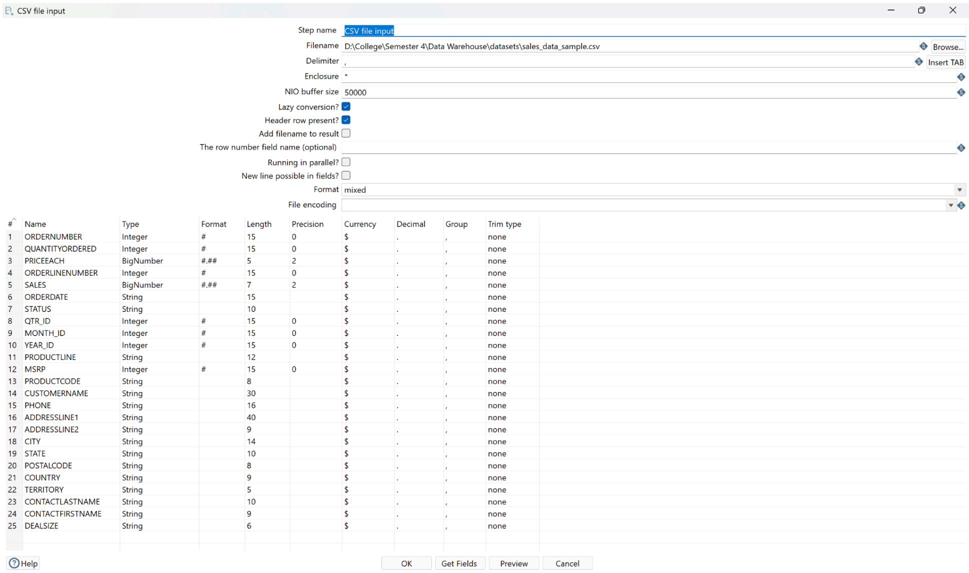
10. Penjelasan Hasil : Hasil ETL menunjukkan bahwa proses transformasi dimDate berhasil dijalankan tanpa error. Proses ini mencakup konversi tipe data kolom date dari String ke tipe Date dengan format M/d/yyyy H:mm melalui konfigurasi Meta-data pada step Select values. Setelah deduplikasi berdasarkan tanggal transaksi, diperoleh 252 baris unik yang berhasil di-insert ke tabel dimdate.

5. Transformasi dim Location

1. Membuat object

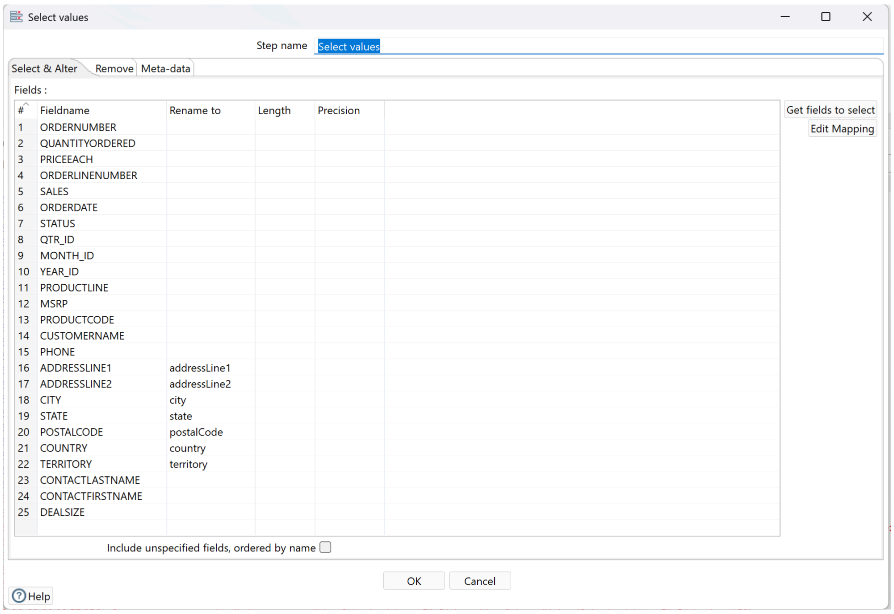


2. Konfigurasi CSV file input

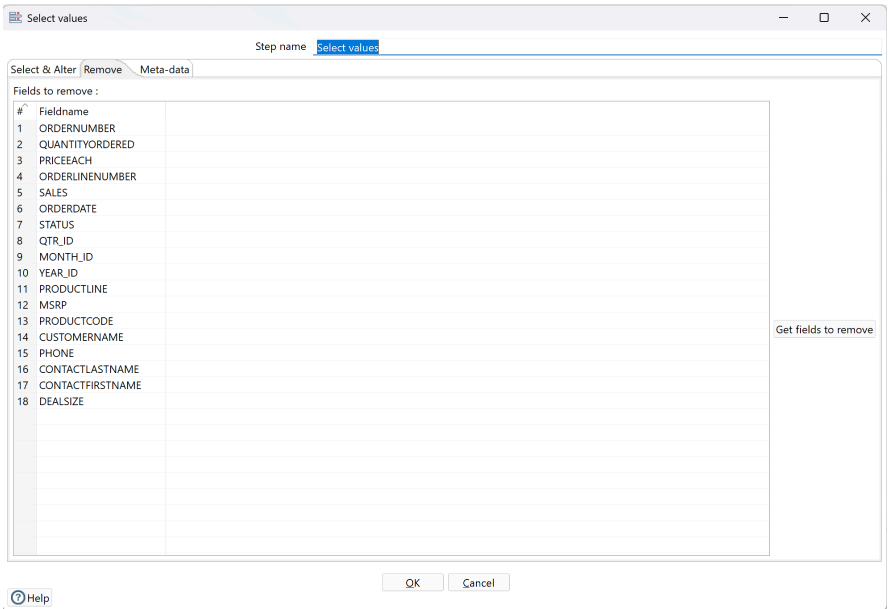


#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	$	.	,	none
2	QUANTITYORDERED	Integer	#	15	0	$	.	,	none
3	PRICEEACH	BigNumber	#.##	5	2	$	.	,	none
4	ORDERLINENUMBER	Integer	#	15	0	$	.	,	none
5	SALES	BigNumber	#.##	7	2	$	.	,	none
6	ORDERDATE	String		15		$	.	,	none
7	STATUS	String		10		$	.	,	none
8	QTR_ID	Integer	#	15	0	$	.	,	none
9	MONTH_ID	Integer	#	15	0	$	.	,	none
10	YEAR_ID	Integer	#	15	0	$	.	,	none
11	PRODUCTLINE	String		12		$	.	,	none
12	MSRP	Integer	#	15	0	$	.	,	none
13	PRODUCTCODE	String		8		$	.	,	none
14	CUSTOMERNAME	String		30		$	.	,	none
15	PHONE	String		16		$	.	,	none
16	ADDRESSLINE1	String		40		$	.	,	none
17	ADDRESSLINE2	String		9		$	.	,	none
18	CITY	String		14		$	.	,	none
19	STATE	String		10		$	.	,	none
20	POSTALCODE	String		8		$	.	,	none
21	COUNTRY	String		9		$	.	,	none
22	TERRITORY	String		5		$	.	,	none
23	CONTACTLASTNAME	String		10		$	.	,	none
24	CONTACTFIRSTNAME	String		9		$	.	,	none
25	DEALSIZE	String		6		$	.	,	none

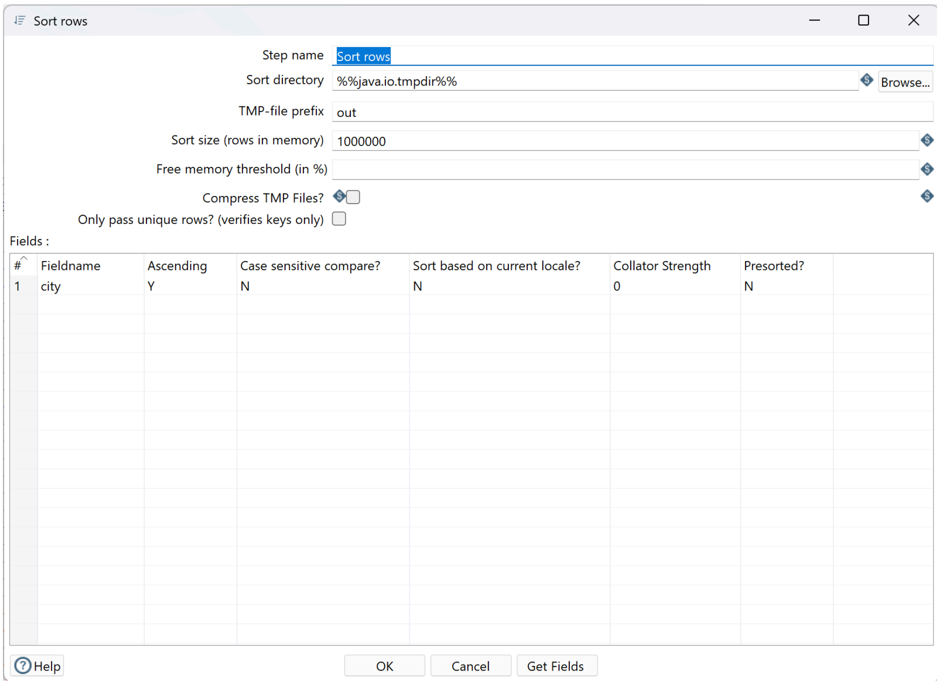
3. Select values
   - Select & alter



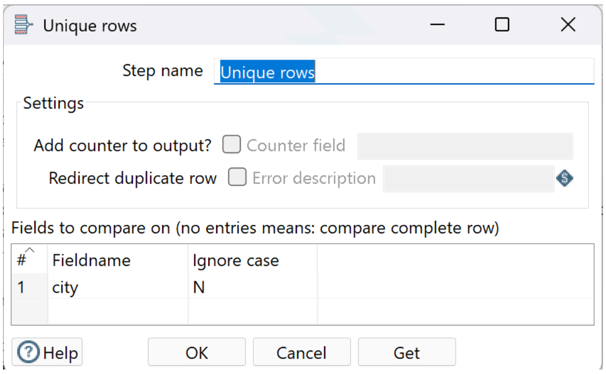
- Remove



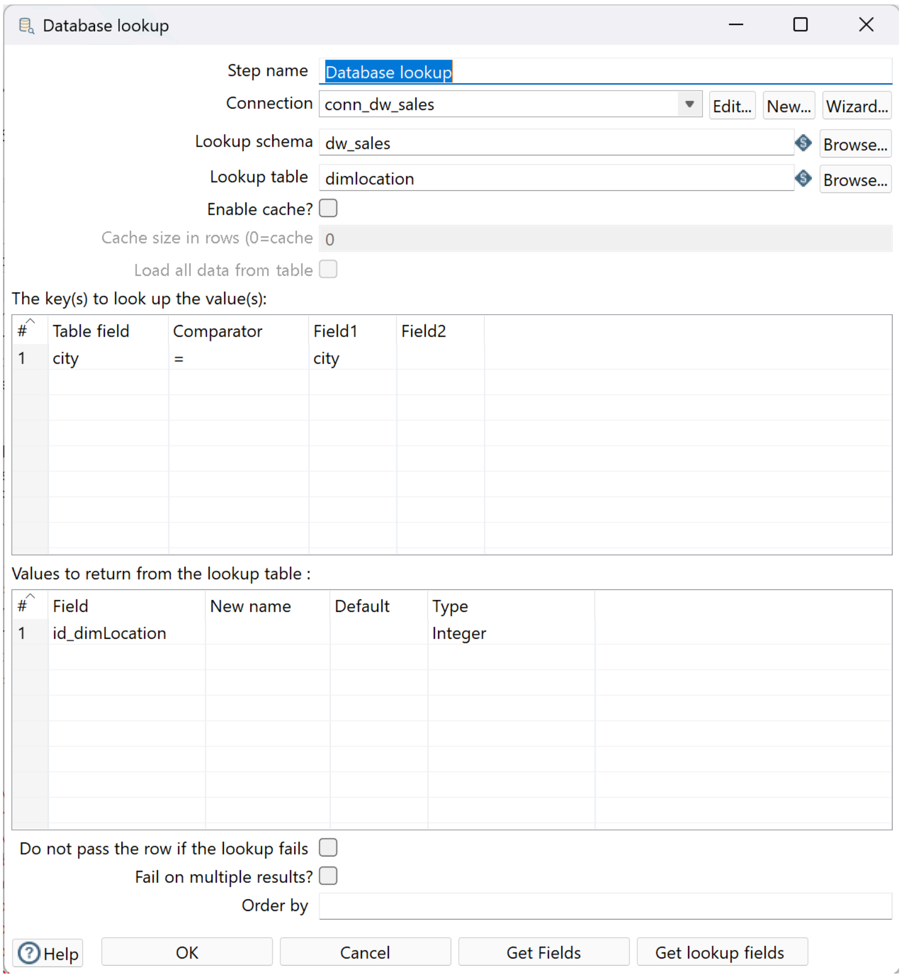
4. Sort rows



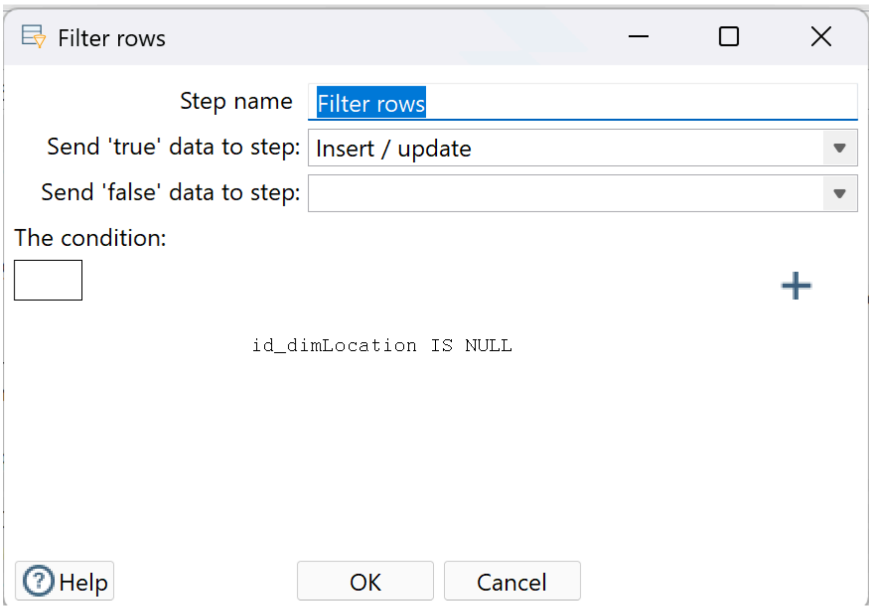
5. Unique rows



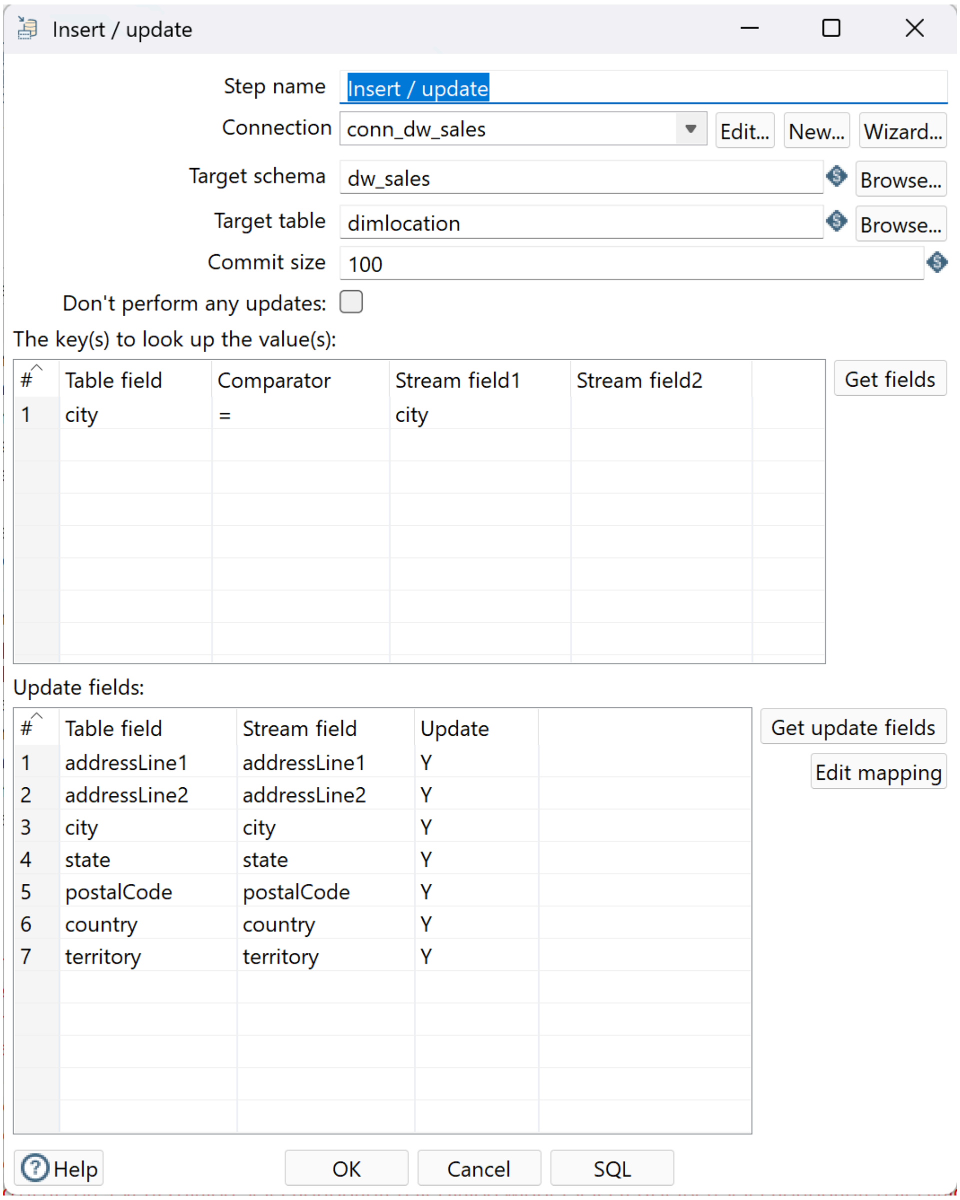
6. Database lookup



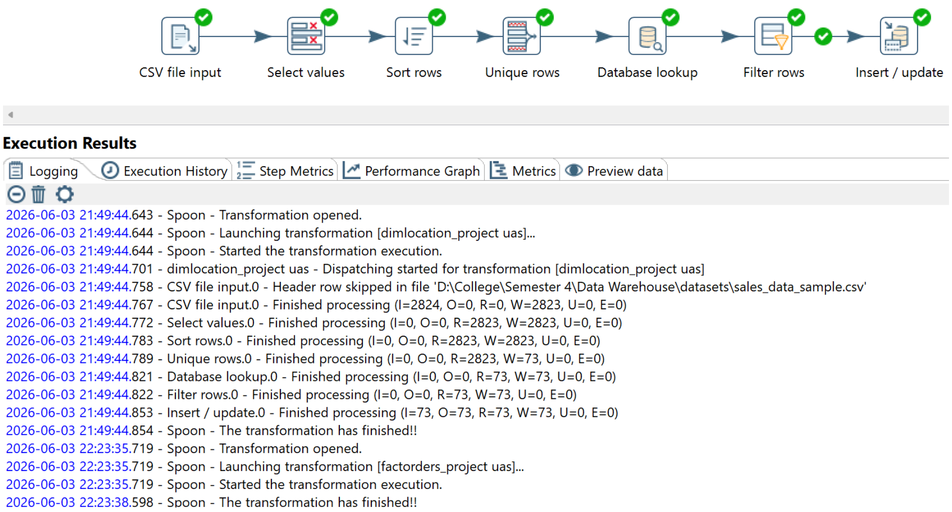
7. Filter rows

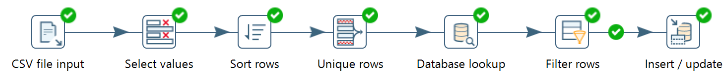


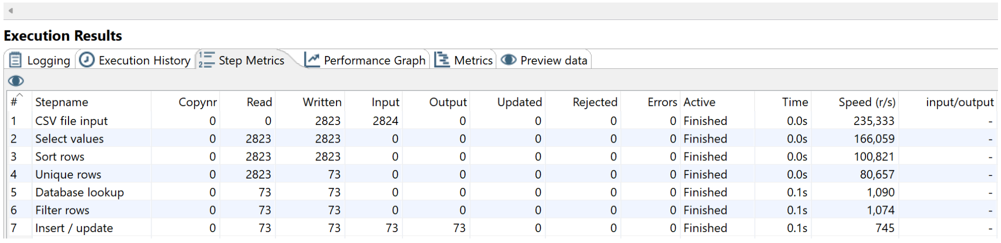
8. Insert / Update


Insert / update
Step name: Insert / update
Connection: conn_dw_sales
Edit... New... Wizard...
Target schema: dw_sales
Browse...
Target table: dimlocation
Browse...
Commit size: 100
Don't perform any updates:
The key(s) to look up the value(s):
| Table field | Comparator | Stream field1 | Stream field2
1 | city | = | city
Get fields
Update fields:
| Table field | Stream field | Update
1 | addressLine1 | addressLine1 | Y
2 | addressLine2 | addressLine2 | Y
3 | city | city | Y
4 | state | state | Y
5 | postalCode | postalCode | Y
6 | country | country | Y
7 | territory | territory | Y
Get update fields
Edit mapping
Help
OK
Cancel
SQL

9. Hasil run


CSV file input
Select values
Sort rows
Unique rows
Database lookup
Filter rows
Insert / update
Execution Results
Logging
Execution History
Step Metrics
Performance Graph
Metrics
Preview data
2026-06-03 21:49:44.643 - Spoon - Transformation opened.
2026-06-03 21:49:44.644 - Spoon - Launching transformation [dimlocation_project uas]...
2026-06-03 21:49:44.644 - Spoon - Started the transformation execution.
2026-06-03 21:49:44.701 - dimlocation_project uas - Dispatching started for transformation [dimlocation_project uas]
2026-06-03 21:49:44.758 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 21:49:44.767 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 21:49:44.772 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 21:49:44.783 - Sort rows.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 21:49:44.789 - Unique rows.0 - Finished processing (I=0, O=0, R=2823, W=73, U=0, E=0)
2026-06-03 21:49:44.821 - Database lookup.0 - Finished processing (I=0, O=0, R=73, W=73, U=0, E=0)
2026-06-03 21:49:44.822 - Filter rows.0 - Finished processing (I=0, O=0, R=73, W=73, U=0, E=0)
2026-06-03 21:49:44.853 - Insert / update.0 - Finished processing (I=73, O=73, R=73, W=73, U=0, E=0)
2026-06-03 21:49:44.854 - Spoon - The transformation has finished!!
2026-06-03 22:23:35.719 - Spoon - Transformation opened.
2026-06-03 22:23:35.719 - Spoon - Launching transformation [factorders_project uas]...
2026-06-03 22:23:35.719 - Spoon - Started the transformation execution.
2026-06-03 22:23:38.598 - Spoon - The transformation has finished!!

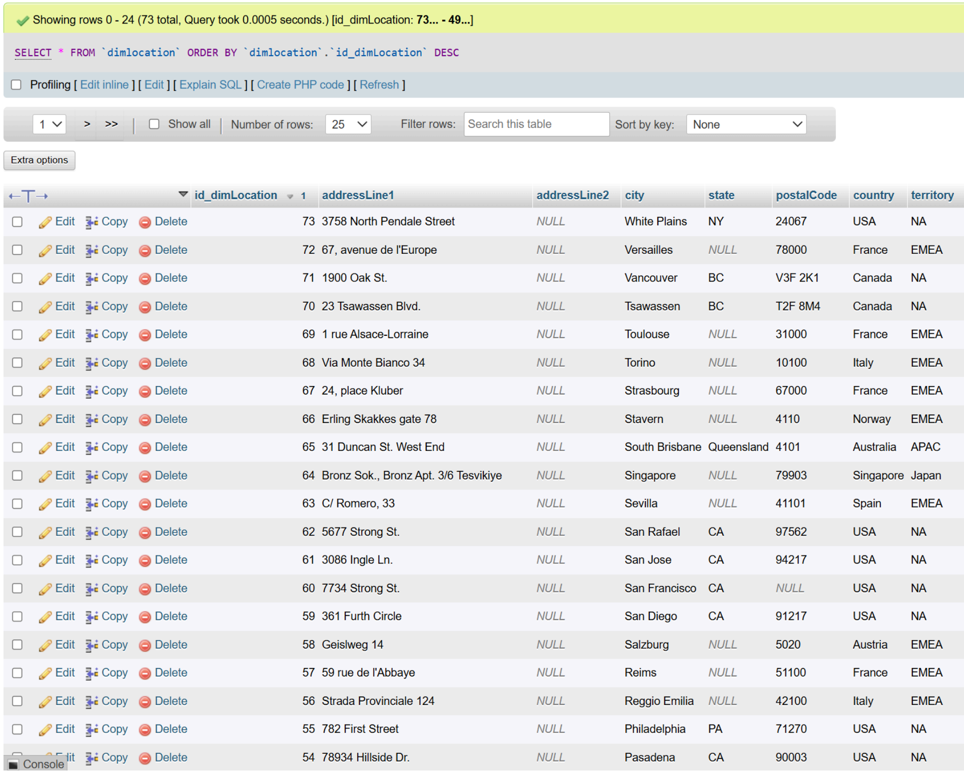




Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	235,333	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.0s	166,059	-
3	Sort rows	0	2823	2823	0	0	0	0	0	Finished	0.0s	100,821	-
4	Unique rows	0	2823	73	0	0	0	0	0	Finished	0.0s	80,657	-
5	Database lookup	0	73	73	0	0	0	0	0	Finished	0.1s	1,090	-
6	Filter rows	0	73	73	0	0	0	0	0	Finished	0.1s	1,074	-
7	Insert / update	0	73	73	73	73	0	0	0	Finished	0.1s	745	-



Showing rows 0 - 24 (73 total, Query took 0.0005 seconds.) [id_dimLocation: **73... - 49...**]

```
SELECT * FROM `dimlocation` ORDER BY `dimlocation`.`id_dimLocation` DESC
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

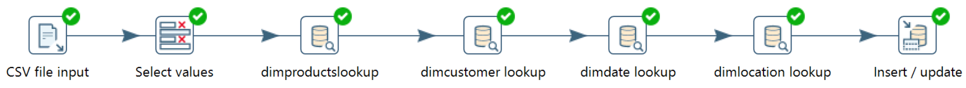
Extra options

←T→	id_dimLocation 1	addressLine1	addressLine2	city	state	postalCode	country	territory
Edit Copy Delete	73	3758 North Pendale Street	NULL	White Plains	NY	24067	USA	NA
Edit Copy Delete	72	67, avenue de l'Europe	NULL	Versailles	NULL	78000	France	EMEA
Edit Copy Delete	71	1900 Oak St.	NULL	Vancouver	BC	V3F 2K1	Canada	NA
Edit Copy Delete	70	23 Tsawassen Blvd.	NULL	Tsawassen	BC	T2F 8M4	Canada	NA
Edit Copy Delete	69	1 rue Alsace-Lorraine	NULL	Toulouse	NULL	31000	France	EMEA
Edit Copy Delete	68	Via Monte Bianco 34	NULL	Torino	NULL	10100	Italy	EMEA
Edit Copy Delete	67	24, place Kluber	NULL	Strasbourg	NULL	67000	France	EMEA
Edit Copy Delete	66	Erling Skakkes gate 78	NULL	Stavern	NULL	4110	Norway	EMEA
Edit Copy Delete	65	31 Duncan St. West End	NULL	South Brisbane	Queensland	4101	Australia	APAC
Edit Copy Delete	64	Bronz Sok., Bronz Apt. 3/6 Tesvikiye	NULL	Singapore	NULL	79903	Singapore	Japan
Edit Copy Delete	63	C/ Romero, 33	NULL	Sevilla	NULL	41101	Spain	EMEA
Edit Copy Delete	62	5677 Strong St.	NULL	San Rafael	CA	97562	USA	NA
Edit Copy Delete	61	3086 Ingle Ln.	NULL	San Jose	CA	94217	USA	NA
Edit Copy Delete	60	7734 Strong St.	NULL	San Francisco	CA	NULL	USA	NA
Edit Copy Delete	59	361 Furth Circle	NULL	San Diego	CA	91217	USA	NA
Edit Copy Delete	58	Geislweg 14	NULL	Salzburg	NULL	5020	Austria	EMEA
Edit Copy Delete	57	59 rue de l'Abbaye	NULL	Reims	NULL	51100	France	EMEA
Edit Copy Delete	56	Strada Provinciale 124	NULL	Reggio Emilia	NULL	42100	Italy	EMEA
Edit Copy Delete	55	782 First Street	NULL	Philadelphia	PA	71270	USA	NA
Console Edit Copy Delete	54	78934 Hillside Dr.	NULL	Pasadena	CA	90003	USA	NA

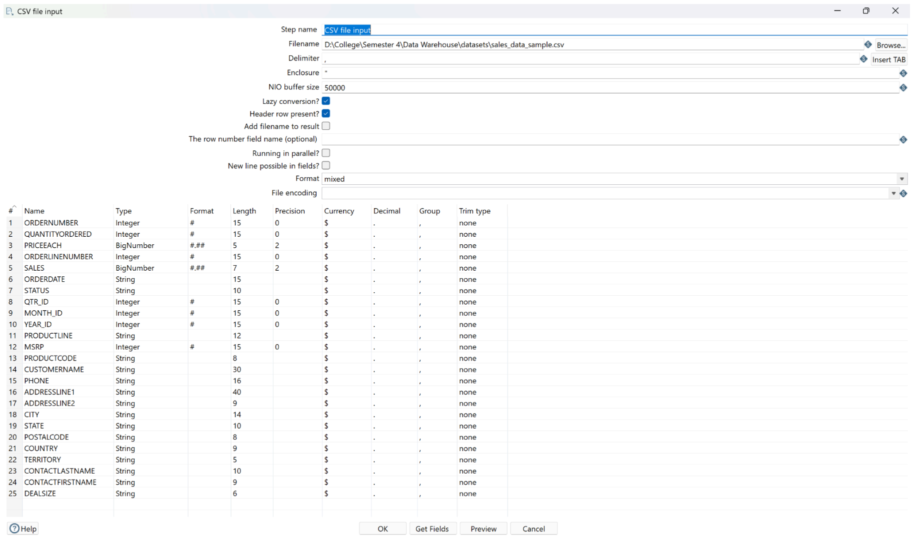
10. Penjelasan Hasil : Hasil ETL menunjukkan bahwa proses transformasi dimLocation berhasil dijalankan tanpa error. Dari 2.823 baris data, setelah deduplikasi berdasarkan city diperoleh 73 baris unik yang berhasil di-insert ke tabel dimlocation. Kolom yang bersifat opsional seperti addressLine2, state, postalCode, dan territory tetap dapat ditangani dengan baik karena skema DDL telah dikonfigurasi untuk menerima nilai NULL pada kolom-kolom tersebut.

6. Transformasi Facts_orders(Tabel Fakta)

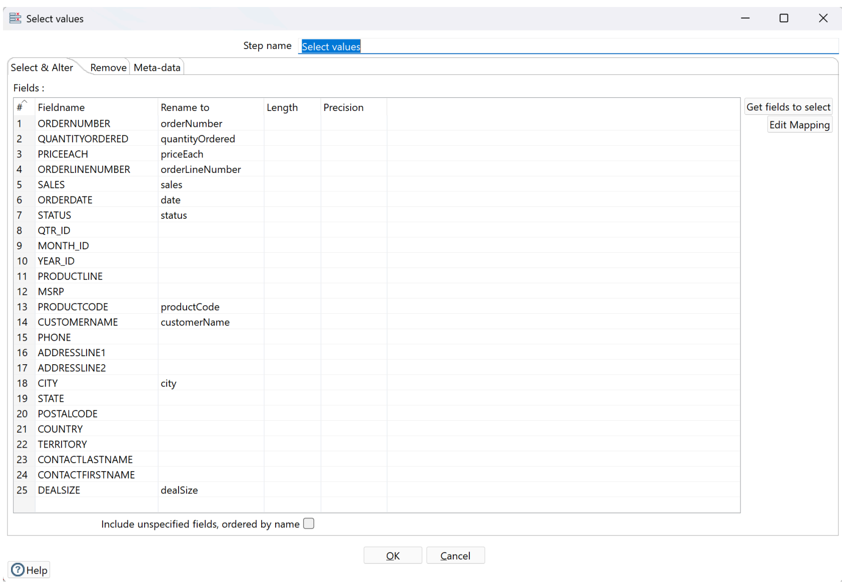
1. Membuat object



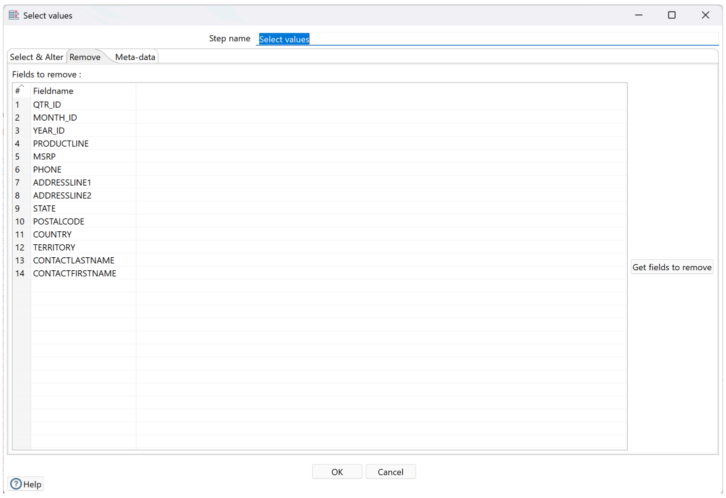
2. Konfigurasi CSV file input



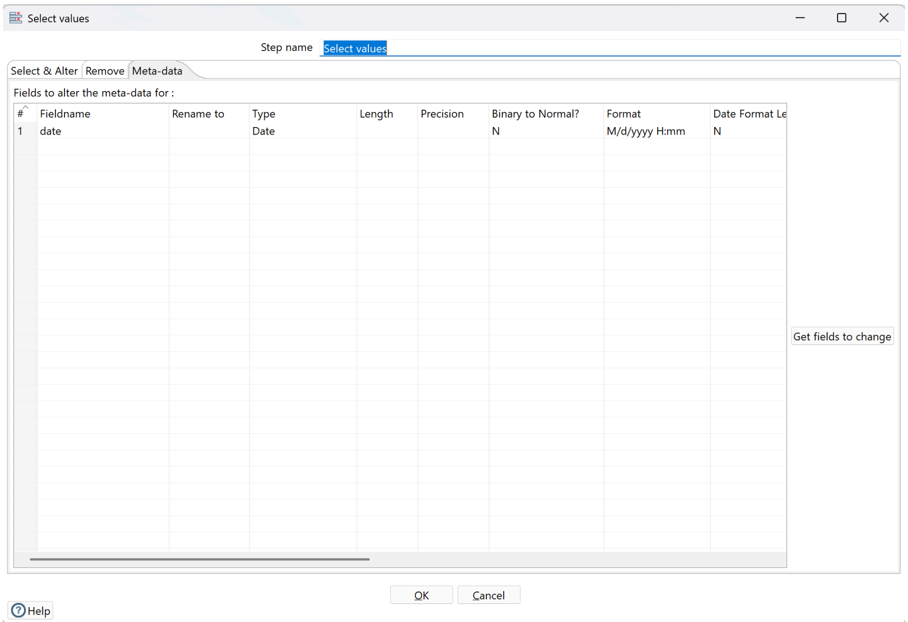
3. Select values
   - Select & alter



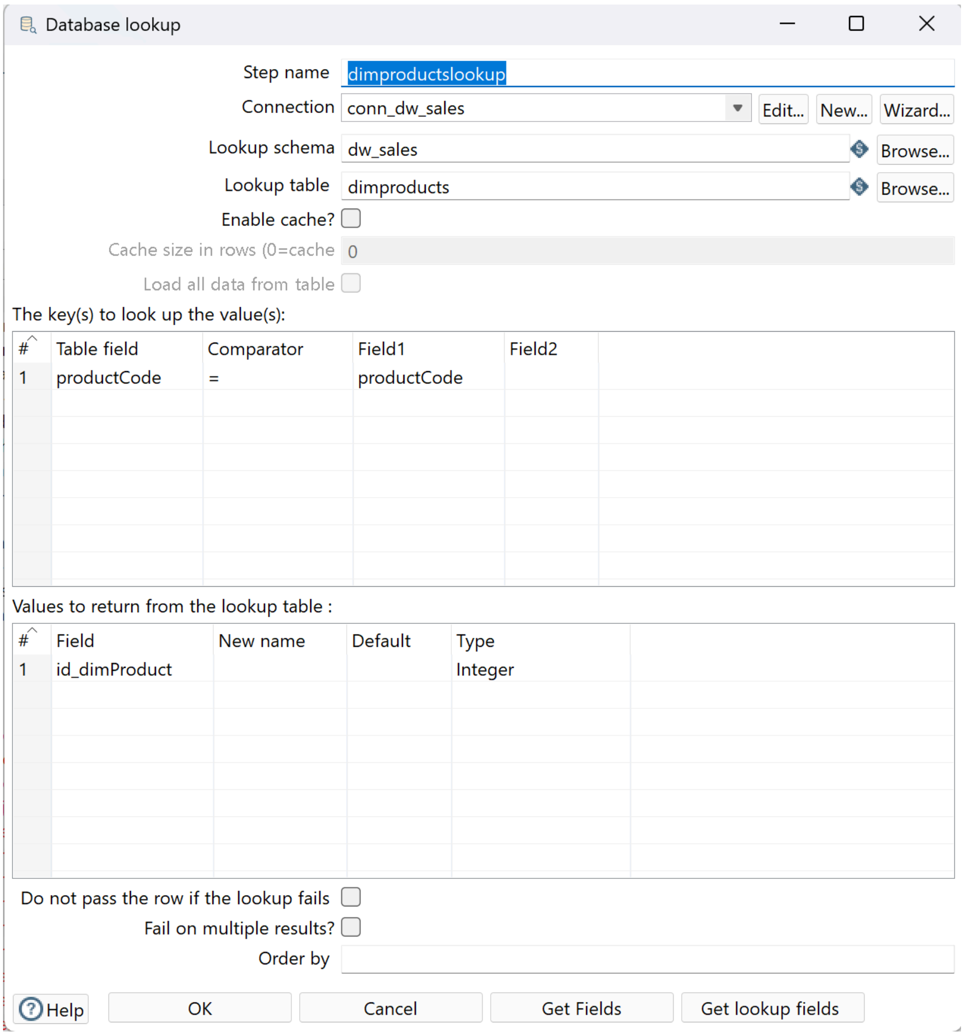
   - Remove



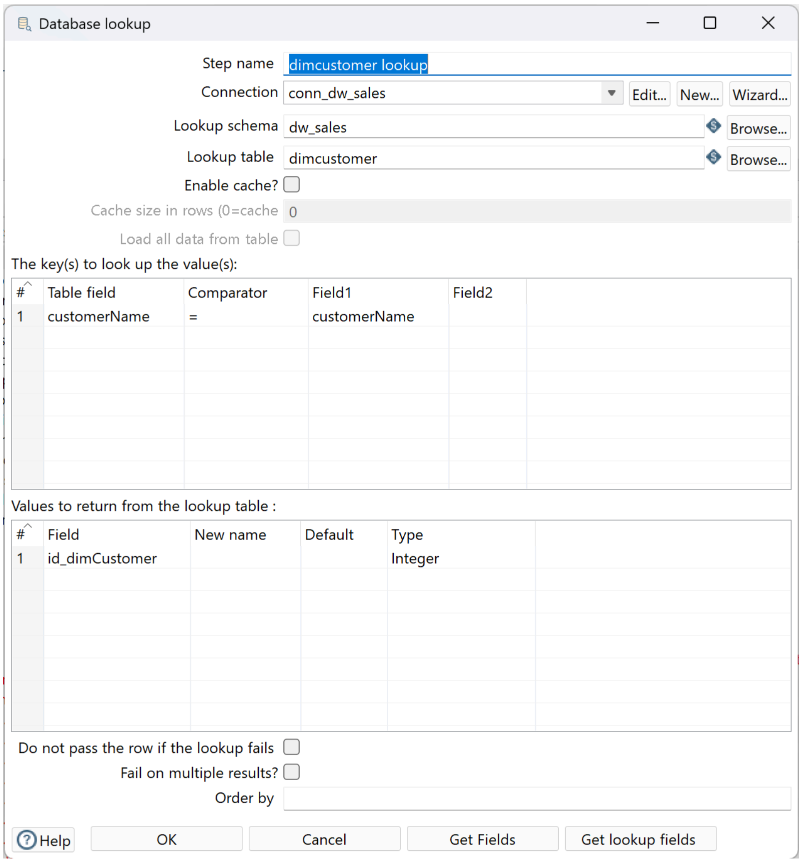
- Meta data



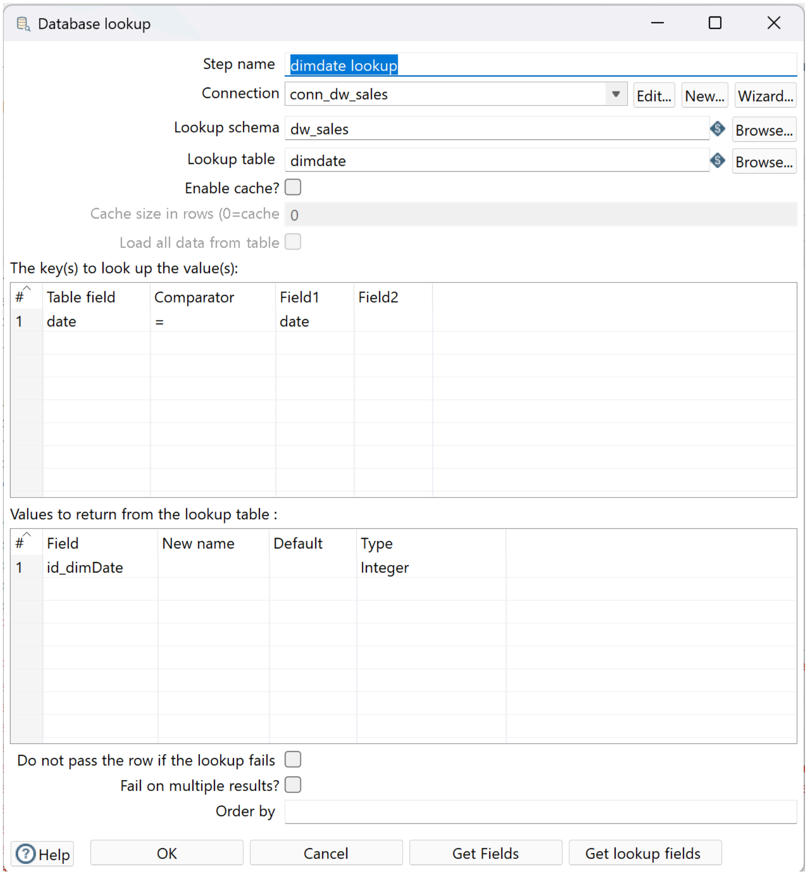
4. dimproductslookup



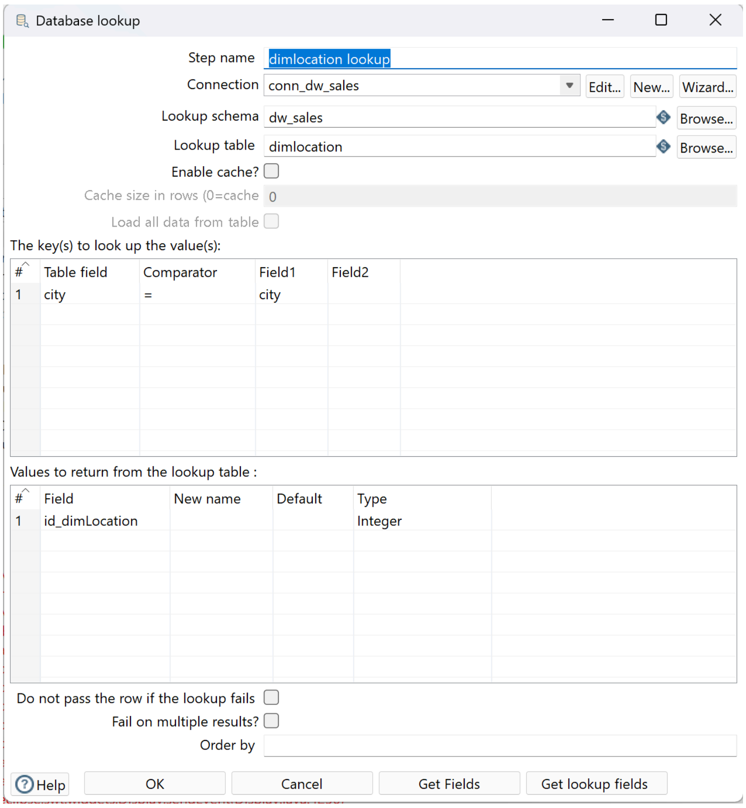
5. dimcustomerlookup



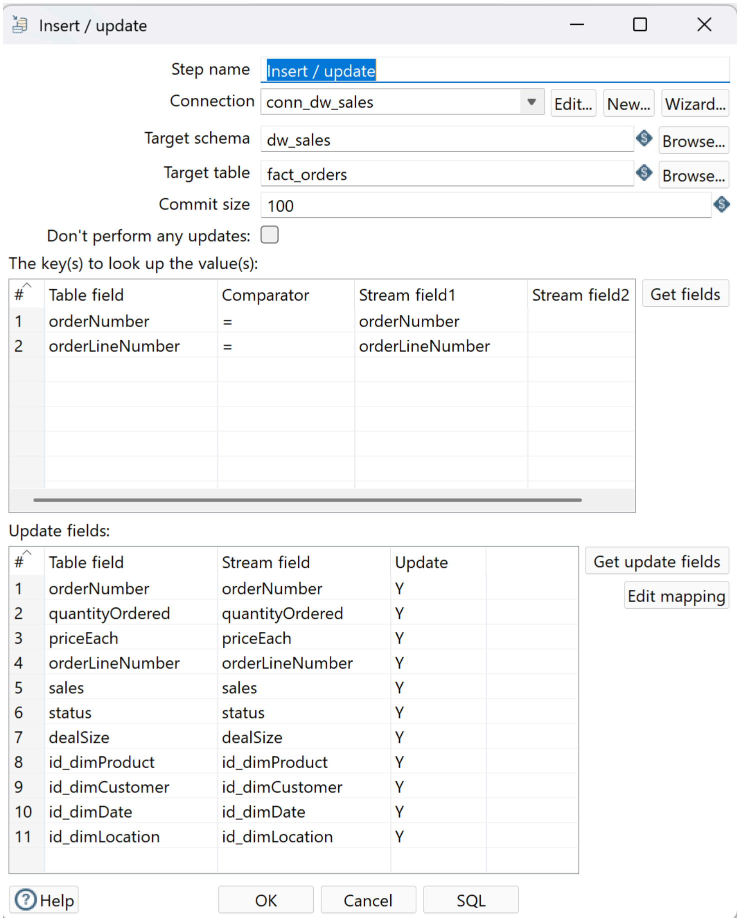
6. dimdatelookup



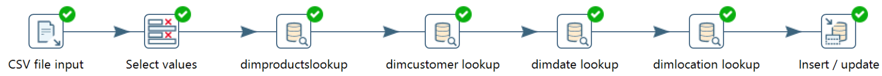
7. dimlocationlookup



8. Insert / Update



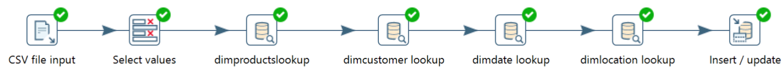
9. Hasil run



Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

```
2026-06-03 22:23:35.719 - Spoon - Transformation opened.
2026-06-03 22:23:35.719 - Spoon - Launching transformation [factorders_project uas]...
2026-06-03 22:23:35.719 - Spoon - Started the transformation execution.
2026-06-03 22:23:35.750 - factorders_project uas - Dispatching started for transformation [factorders_project uas]
2026-06-03 22:23:35.830 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 22:23:35.838 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 22:23:35.844 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:36.840 - dimproductslookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:36.887 - dimcustomer lookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:36.951 - dimdate lookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:37.000 - dimlocation lookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:38.596 - Insert / update.0 - Finished processing (I=2823, O=2823, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:38.598 - Spoon - The transformation has finished!!
```



Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	282,400	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.0s	188,200	-
3	dimproductslookup	0	2823	2823	2823	0	0	0	0	Finished	1.0s	2,790	-
4	dimcustomer lookup	0	2823	2823	2823	0	0	0	0	Finished	1.1s	2,666	-
5	dimdate lookup	0	2823	2823	2823	0	0	0	0	Finished	1.1s	2,514	-
6	dimlocation lookup	0	2823	2823	2823	0	0	0	0	Finished	1.2s	2,409	-
7	Insert / update	0	2823	2823	2823	2823	0	0	0	Finished	2.8s	1,020	-

Showing rows 0 - 24 (2823 total, Query took 0.0004 seconds.) [id_factOrder: 2823... - 2799...]

```
SELECT * FROM `fact_orders` ORDER BY `fact_orders`.`id_factOrder` DESC
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

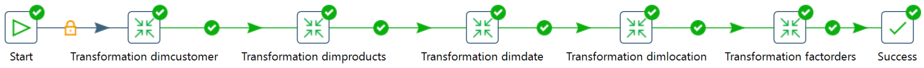
	id_factOrder	orderNumber	id_dimProduct	id_dimCustomer	id_dimDate	id_dimLocation	quantityOrdered	priceEach	orderLineNumber	sales	status	dealSize
Edit Copy Delete	2823	10414	109	38	245	7	47	65.52	9	3079.44	On Hold	Medium
Edit Copy Delete	2822	10397	109	1	232	69	34	62.24	1	2116.16	Shipped	Small
Edit Copy Delete	2821	10386	109	34	223	35	43	100.00	4	5417.57	Resolved	Medium
Edit Copy Delete	2820	10373	109	65	212	52	29	100.00	1	3978.51	Shipped	Medium
Edit Copy Delete	2819	10350	109	34	195	35	20	100.00	15	2244.40	Shipped	Small
Edit Copy Delete	2818	10337	109	20	188	49	42	97.16	5	4080.72	Shipped	Medium
Edit Copy Delete	2817	10327	109	28	180	26	37	86.74	4	3209.38	Resolved	Medium
Edit Copy Delete	2816	10315	109	46	173	43	40	55.69	5	2227.60	Shipped	Small
Edit Copy Delete	2815	10306	109	11	168	37	35	59.51	6	2082.85	Shipped	Small
Edit Copy Delete	2814	10293	109	2	159	68	32	60.06	1	1921.92	Shipped	Small
Edit Copy Delete	2813	10283	109	70	150	70	33	51.32	12	1693.56	Shipped	Small
Edit Copy Delete	2812	10273	109	66	142	11	37	45.86	10	1696.82	Shipped	Small
Edit Copy Delete	2811	10261	109	67	132	41	29	50.78	7	1472.62	Shipped	Small
Edit Copy Delete	2810	10248	109	47	122	49	23	65.52	9	1506.96	Cancelled	Small
Edit Copy Delete	2809	10232	109	39	109	17	24	49.69	3	1192.56	Shipped	Small
Edit Copy Delete	2808	10222	109	22	99	59	36	63.34	18	2280.24	Shipped	Small
Edit Copy Delete	2807	10208	109	73	86	34	42	63.88	6	2682.96	Shipped	Small
Edit Copy Delete	2806	10197	109	33	79	3	42	50.23	12	2109.66	Shipped	Small
Edit Copy Delete	2805	10185	109	56	73	45	28	64.43	6	1804.04	Shipped	Small
Edit Copy Delete	2804	10177	109	16	68	35	40	50.23	6	2009.20	Shipped	Small

Console

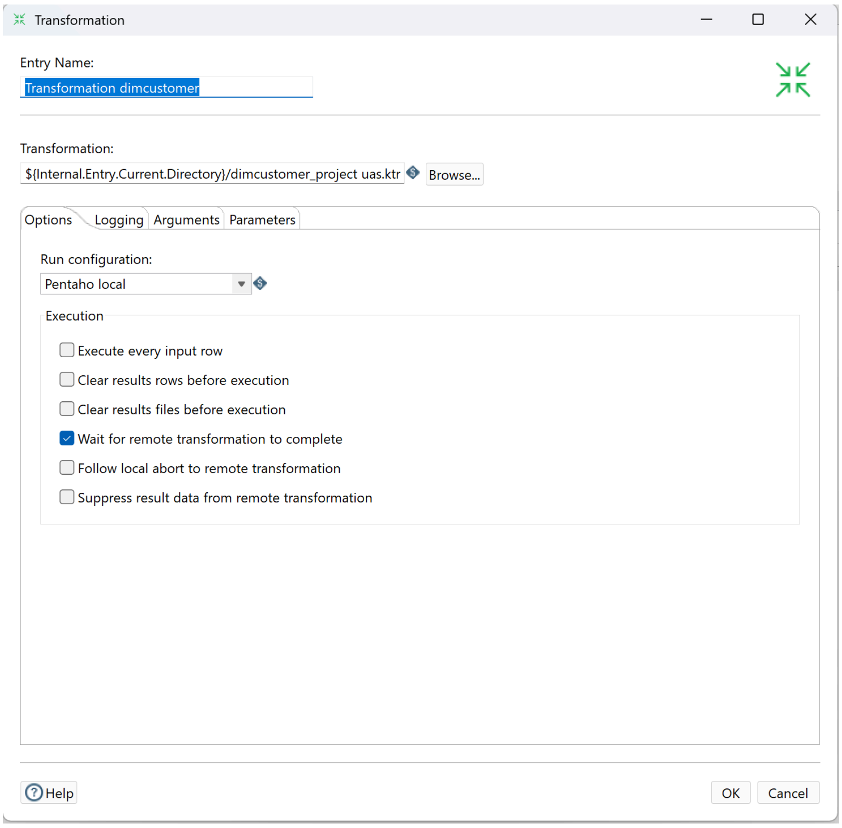
10. Penjelasan Hasil : Hasil ETL menunjukkan bahwa proses transformasi fact_orders berhasil dijalankan tanpa error. Seluruh 2.823 baris data transaksi berhasil di-insert ke tabel fact_orders setelah melalui proses multi-lookup ke keempat tabel dimensi untuk mendapatkan surrogate key id_dimProduct, id_dimCustomer, id_dimDate, dan id_dimLocation. Hal ini membuktikan bahwa seluruh relasi foreign key berhasil terisi dengan benar dan integritas referensial antar tabel terjaga.

7. Job

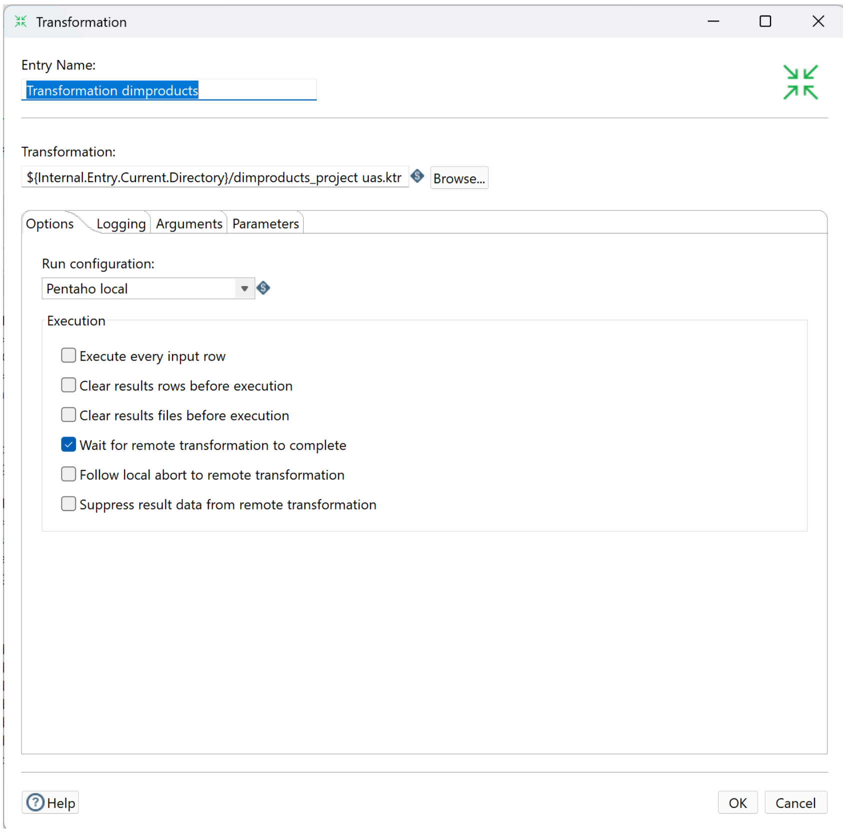
1. Membuat object



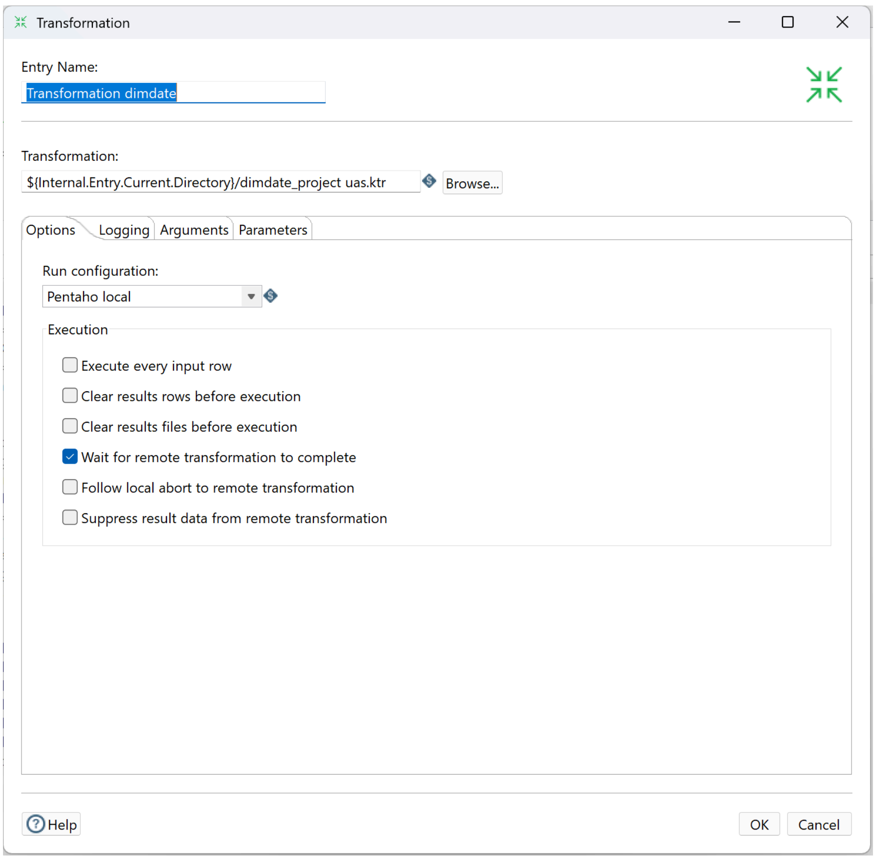
2. Konfigurasi transformation dimcustomer



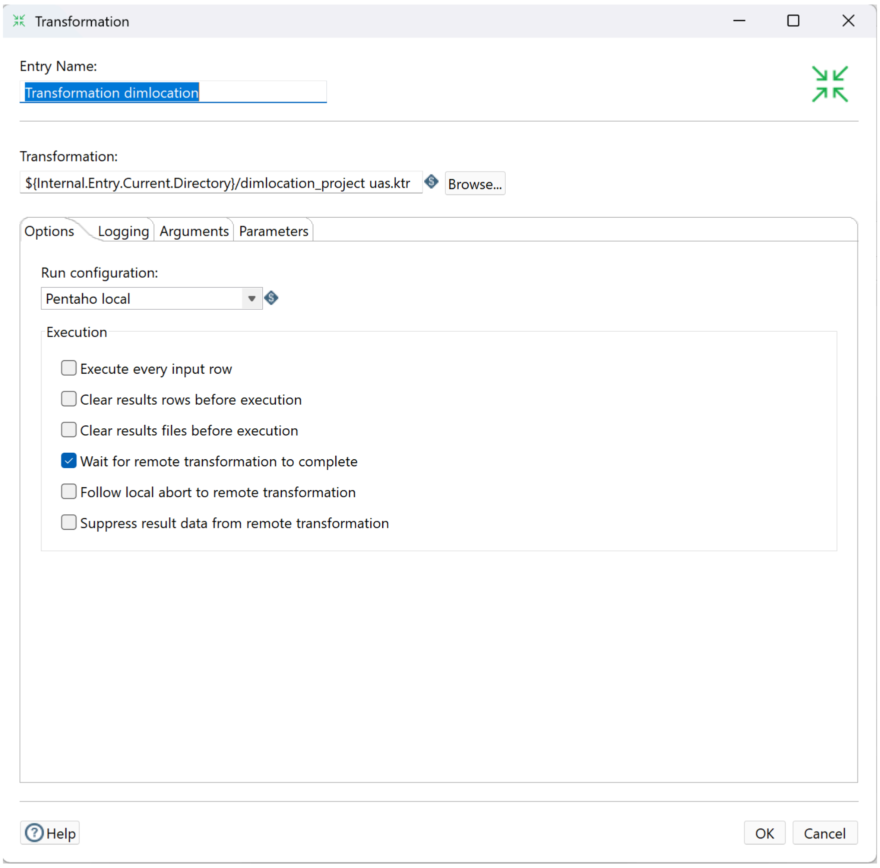
3. Konfigurasi transformation dimproducts



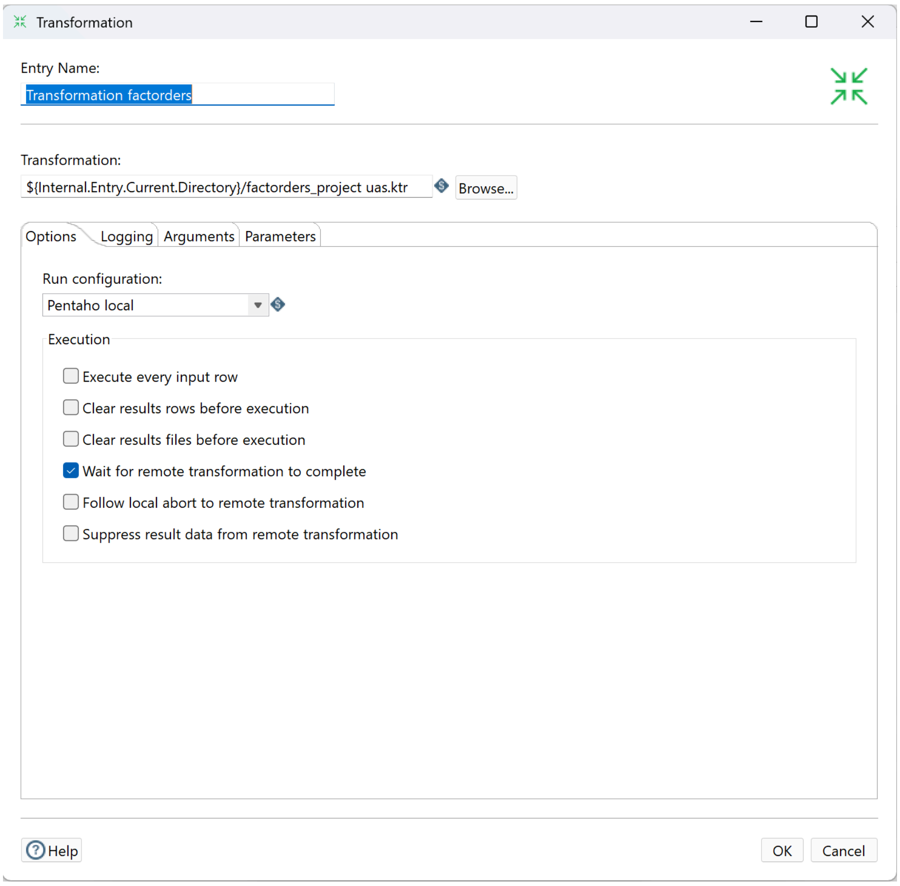
4. Konfigurasi transformation dimdate



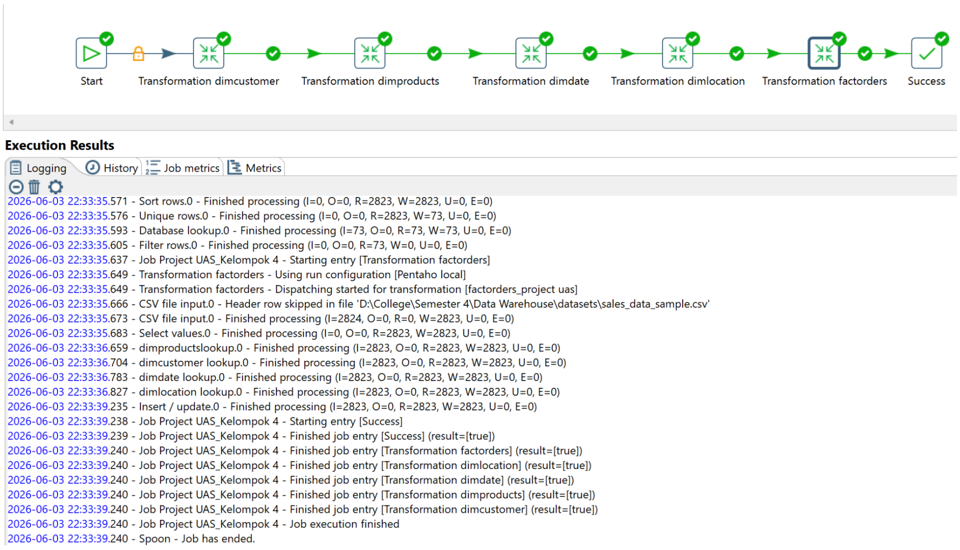
5. Konfigurasi transformation dimlocation

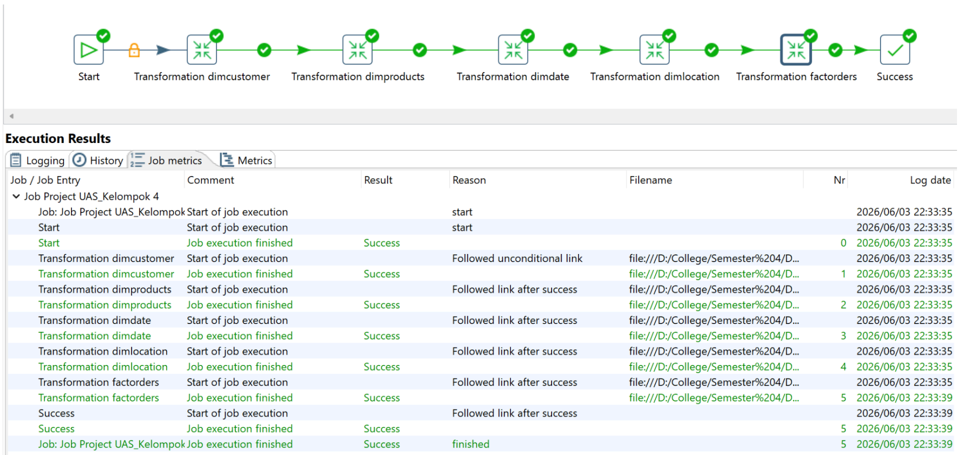


6. Konfigurasi transformation factorders



7. Hasil run





Job / Job Entry	Comment	Result	Reason	Filename	Nr	Log date
Job Project UAS_Kelompok 4						
Job: Job Project UAS_Kelompok	Start of job execution		start			2026/06/03 22:33:35
Start	Start of job execution		start			2026/06/03 22:33:35
Start	Job execution finished	Success			0	2026/06/03 22:33:35
Transformation dimcustomer	Start of job execution		Followed unconditional link	file:///D:/College/Semester%204/D...		2026/06/03 22:33:35
Transformation dimcustomer	Job execution finished	Success		file:///D:/College/Semester%204/D...	1	2026/06/03 22:33:35
Transformation dimproducts	Start of job execution		Followed link after success	file:///D:/College/Semester%204/D...		2026/06/03 22:33:35
Transformation dimproducts	Job execution finished	Success		file:///D:/College/Semester%204/D...	2	2026/06/03 22:33:35
Transformation dimdate	Start of job execution		Followed link after success	file:///D:/College/Semester%204/D...		2026/06/03 22:33:35
Transformation dimdate	Job execution finished	Success		file:///D:/College/Semester%204/D...	3	2026/06/03 22:33:35
Transformation dimlocation	Start of job execution		Followed link after success	file:///D:/College/Semester%204/D...		2026/06/03 22:33:35
Transformation dimlocation	Job execution finished	Success		file:///D:/College/Semester%204/D...	4	2026/06/03 22:33:35
Transformation factorders	Start of job execution		Followed link after success	file:///D:/College/Semester%204/D...		2026/06/03 22:33:35
Transformation factorders	Job execution finished	Success		file:///D:/College/Semester%204/D...	5	2026/06/03 22:33:39
Success	Start of job execution		Followed link after success			2026/06/03 22:33:39
Success	Job execution finished	Success			5	2026/06/03 22:33:39
Job: Job Project UAS_Kelompok	Job execution finished	Success	finished		5	2026/06/03 22:33:39

8. Penjelasan Hasil : Hasil eksekusi Job menunjukkan bahwa seluruh transformation berhasil dijalankan secara berurutan tanpa error dengan status Success pada setiap entry. Urutan eksekusi yang dimulai dari dimcustomer, dimproducts, dimdate, dimlocation, hingga factorders memastikan bahwa seluruh tabel dimensi telah terisi terlebih dahulu sebelum tabel fakta diproses, sehingga tidak terjadi pelanggaran referential integrity selama proses load berlangsung.

E. Tahap Load (Pemuatan Data)

Setelah proses transformasi selesai, data dimuat ke dalam database dw_sales menggunakan step Insert/Update di Pentaho Spoon. Proses load dilakukan dengan memperhatikan integritas data, kunci primer, dan relasi antar tabel. Tabel dimensi dimuat terlebih dahulu sebelum tabel fakta untuk memastikan seluruh foreign key pada fact_orders sudah tersedia di tabel dimensi yang bersesuaian, sehingga tidak terjadi pelanggaran referential integrity. Setiap tabel dimensi menggunakan primary key AUTO_INCREMENT yang di-generate otomatis oleh MySQL, sedangkan tabel fakta menggunakan foreign key hasil lookup dari masing-masing dimensi.

1. Load dimcustomer

   Data pelanggan berhasil dimuat sebanyak 92 baris ke tabel dimcustomer. Proses ini melibatkan penghapusan duplikat berdasarkan customerName sehingga setiap pelanggan hanya tersimpan satu kali. Tidak ditemukan error pada proses load ini.

2. Load dimproducts

   Data produk berhasil dimuat sebanyak 109 baris ke tabel dimproducts. Duplikat berdasarkan productCode berhasil dieliminasi sehingga setiap produk hanya tercatat satu kali dalam dimensi produk.

3. Load dimdate

   Data waktu berhasil dimuat sebanyak 252 baris ke tabel dimdate. Kolom date berhasil dikonversi dari format String ke tipe Date dengan format M/d/yyyy H:mm sebelum dimuat ke database.

4. Load dimlocation

   Data lokasi berhasil dimuat sebanyak 73 baris ke tabel dimlocation. Duplikat berdasarkan city berhasil dieliminasi sehingga setiap lokasi hanya tersimpan satu kali.

5. Load fact_orders

   Data transaksi berhasil dimuat sebanyak 2.823 baris ke tabel fact_orders. Proses ini melibatkan lookup ke seluruh tabel dimensi untuk mendapatkan foreign key yang sesuai, yaitu id_dimProduct, id_dimCustomer, id_dimDate, dan id_dimLocation, sebelum data dimasukkan ke tabel fakta.

6. Eksekusi Job

   Seluruh transformation dijalankan secara berurutan melalui satu Job di Pentaho Spoon. Job berhasil dieksekusi tanpa error dengan urutan: dimcustomer_project uas → dimproducts_project uas → dimdate_project uas → dimlocation_project uas → factorders_project uas. Seluruh data dari file CSV sales_data_sample.csv yang terdiri dari 2.823 baris transaksi berhasil dimuat ke dalam data warehouse dw_sales dengan struktur star schema yang telah dirancang.

IV. HASIL DAN ANALISIS KEY PERFORMANCE INDICATORS (KPI)

A. Hasil Implementasi Data Warehouse

Setelah proses ETL selesai dijalankan, data warehouse GlobalScale Retail berhasil dibangun dengan struktur star schema yang terdiri dari satu tabel fakta dan empat tabel dimensi. Data yang semula tersebar dalam satu file CSV flat berhasil dinormalisasi ke dalam struktur relasional yang terpisah, siap untuk digunakan dalam analisis multidimensional.

Keberhasilan implementasi ditunjukkan dengan terpenuhinya integritas referensial antar tabel, tidak adanya nilai null pada kolom kunci, serta konsistensi jumlah record antara sumber data dan data warehouse. Dari dataset awal yang memiliki 2.823 baris transaksi, seluruh data berhasil dimuat ke dalam tabel fakta tanpa kehilangan data.

B. Analisis Key Performance Indicators (KPI)

Data warehouse yang telah dibangun dapat dimanfaatkan untuk menghasilkan berbagai laporan analitik bisnis. Berikut adalah tiga KPI utama yang dapat dihasilkan beserta contoh query SQL yang digunakan:

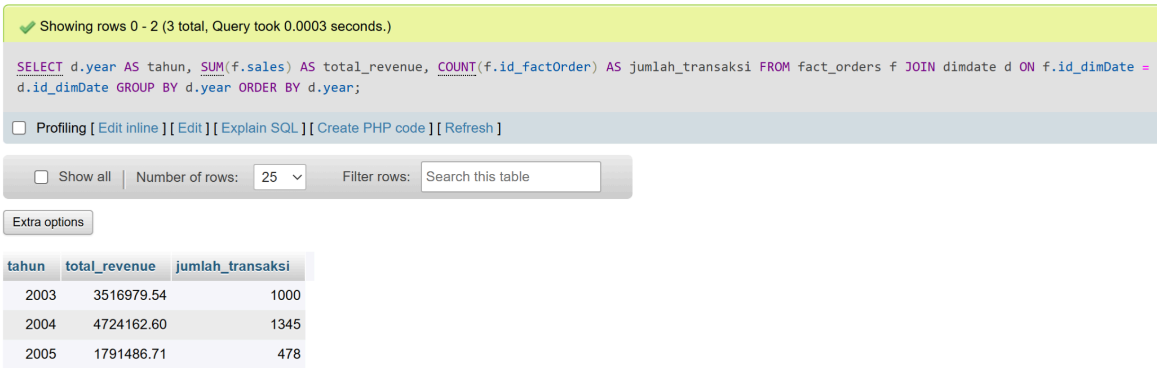
KPI 1: Total Revenue per Tahun

KPI ini mengukur total pendapatan (revenue) perusahaan secara keseluruhan di setiap tahun. Dari data terlihat bahwa 2004 adalah tahun dengan revenue tertinggi ($4,72 juta), naik sekitar 34% dari 2003. Revenue 2005 tampak rendah ($1,79 juta) karena data hanya mencakup sampai pertengahan 2005 (tidak full year). Tren 2003→2004 menunjukkan pertumbuhan bisnis yang signifikan.

Query :

```
SELECT
    d.year AS tahun,
    SUM(f.sales) AS total_revenue,
    COUNT(f.id_factOrder) AS jumlah_transaksi
FROM fact_orders f
JOIN dimdate d ON f.id_dimDate = d.id_dimDate
GROUP BY d.year
ORDER BY d.year;
```

Hasil :



tahun	total_revenue	jumlah_transaksi
2003	3516979.54	1000
2004	4724162.60	1345
2005	1791486.71	478

Hasil query KPI Total Revenue per Tahun menunjukkan bahwa data warehouse berhasil menghasilkan informasi agregat penjualan berdasarkan dimensi waktu. Dari hasil tersebut terlihat bahwa tahun 2004 mencatatkan revenue tertinggi sebesar $4.724.162,60 dengan 1.345 transaksi, naik sekitar 34% dibandingkan tahun 2003 yang mencapai $3.516.979,54. Sementara revenue tahun 2005 tampak lebih rendah di angka $1.791.486,71 karena data hanya mencakup sebagian tahun (tidak full year). KPI ini bermanfaat bagi manajemen untuk memantau tren pertumbuhan bisnis dari tahun ke tahun serta menjadi dasar perencanaan target penjualan periode berikutnya.

KPI 2: Product Line dengan Revenue Tertinggi

KPI ini mengidentifikasi lini produk mana yang paling menguntungkan. Classic Cars mendominasi dengan revenue $3,9 juta — hampir 2x lipat dibanding Vintage Cars di posisi kedua. Sementara Trains adalah kontributor terkecil dengan hanya $226K. Insight ini berguna untuk keputusan manajemen seperti alokasi stok, strategi promosi, dan prioritas pengadaan produk.

Query :

```
SELECT
    p.productLine AS kategori_produk,
    SUM(f.sales) AS total_revenue,
    SUM(f.quantityOrdered) AS total_unit_terjual,
    ROUND(AVG(f.priceEach), 2) AS rata_rata_harga
FROM fact_orders f
JOIN dimproducts p ON f.id_dimProduct = p.id_dimProduct
GROUP BY p.productLine
ORDER BY total_revenue DESC;
```

Hasil :

Showing rows 0 - 6 (7 total, Query took 0.0092 seconds.)

```
SELECT p.productLine AS kategori_produk, SUM(f.sales) AS total_revenue, SUM(f.quantityOrdered) AS total_unit_terjual, ROUND(AVG(f.priceEach), 2) AS rata_rata_harga FROM fact_orders f JOIN dimproducts p ON f.id_dimProduct = p.id_dimProduct GROUP BY p.productLine ORDER BY total_revenue DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

kategori_produk	total_revenue 1	total_unit_terjual	rata_rata_harga
Classic Cars	3919615.66	33992	87.34
Vintage Cars	1903150.84	21069	78.15
Motorcycles	1166388.34	11663	83.00
Trucks and Buses	1127789.84	10777	87.53
Planes	975003.57	10727	81.74
Ships	714437.13	8127	83.86
Trains	226243.47	2712	75.65

Hasil query KPI Product Line menunjukkan bahwa data warehouse mampu mengidentifikasi performa penjualan berdasarkan kategori produk. Classic Cars mendominasi dengan total revenue sebesar $3.919.615,66 dan 33.992 unit terjual, hampir dua kali lipat dibandingkan Vintage Cars di posisi kedua. Di sisi lain, Trains menjadi kontributor terkecil dengan revenue hanya $226.243,47. KPI ini bermanfaat bagi manajemen dalam pengambilan keputusan terkait alokasi stok, strategi promosi, serta prioritas pengadaan produk agar sumber daya dapat difokuskan pada lini produk yang paling menguntungkan.

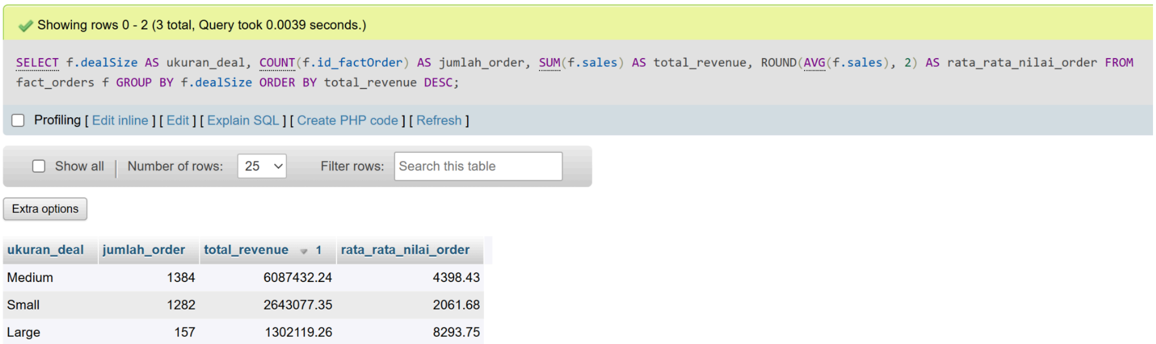
KPI 3: Deal Size yang Paling Banyak Menghasilkan Revenue

KPI ini menganalisis segmentasi transaksi berdasarkan ukuran deal. Menariknya, Medium Deal adalah kontributor terbesar ($6,08 juta / ~61% dari total revenue), bukan Large Deal. Hal ini menunjukkan bahwa bisnis perusahaan lebih banyak ditopang oleh transaksi menengah yang konsisten dan banyak jumlahnya, bukan dari sedikit transaksi bernilai besar. Ini penting untuk strategi penetapan harga dan segmentasi pelanggan.

Query :

```
SELECT
    f.dealSize AS ukuran_deal,
    COUNT(f.id_factOrder) AS jumlah_order,
    SUM(f.sales) AS total_revenue,
    ROUND(AVG(f.sales), 2) AS rata_rata_nilai_order
FROM fact_orders f
GROUP BY f.dealSize
ORDER BY total_revenue DESC;
```

Hasil :



Showing rows 0 - 2 (3 total, Query took 0.0039 seconds.)

```
SELECT f.dealSize AS ukuran_deal, COUNT(f.id_factOrder) AS jumlah_order, SUM(f.sales) AS total_revenue, ROUND(AVG(f.sales), 2) AS rata_rata_nilai_order FROM fact_orders f GROUP BY f.dealSize ORDER BY total_revenue DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

ukuran_deal	jumlah_order	total_revenue 1	rata_rata_nilai_order
Medium	1384	6087432.24	4398.43
Small	1282	2643077.35	2061.68
Large	157	1302119.26	8293.75

Hasil query KPI Deal Size menunjukkan bahwa data warehouse berhasil melakukan segmentasi transaksi berdasarkan ukuran deal. Dari hasil tersebut, Medium Deal menjadi kontributor terbesar dengan total revenue $6.087.432,24 dari 1.384 transaksi, menyumbang sekitar 61% dari total revenue keseluruhan, jauh melampaui Small Deal maupun Large Deal. Hal ini mengindikasikan bahwa bisnis perusahaan lebih banyak ditopang oleh transaksi menengah yang konsisten dan volume tinggi, bukan dari transaksi bernilai sangat besar. KPI ini bermanfaat sebagai acuan dalam strategi penetapan harga dan segmentasi pelanggan agar perusahaan dapat memaksimalkan potensi segmen transaksi yang paling dominan.

KPI 4: Top 10 Pelanggan dengan Pembelian Terbanyak

KPI ini mengidentifikasi pelanggan paling bernilai (high-value customers). Euro Shopping Channel adalah pelanggan terbesar dengan $912K — hampir 1,4x lipat dari pelanggan kedua. Dua pelanggan teratas (Euro Shopping Channel + Mini Gifts Distributors) menyumbang sekitar 16% dari total revenue keseluruhan, menunjukkan ketergantungan yang cukup tinggi pada pelanggan kunci. Perusahaan perlu menjaga loyalitas kedua pelanggan ini sebagai prioritas.

Query :

SELECT

```
    c.customerName AS nama_pelanggan,
    COUNT(f.id_factOrder) AS jumlah_transaksi,
    SUM(f.quantityOrdered) AS total_unit_dibeli,
    SUM(f.sales) AS total_belanja
FROM fact_orders f
JOIN dimcustomer c ON f.id_dimCustomer = c.id_dimCustomer
GROUP BY c.customerName
ORDER BY total_belanja DESC
LIMIT 10;
```

Hasil :

Showing rows 0 - 9 (10 total, Query took 0.0071 seconds.)

```
SELECT c.customerName AS nama_pelanggan, COUNT(f.id_factOrder) AS jumlah_transaksi, SUM(f.quantityOrdered) AS total_unit_dibeli, SUM(f.sales) AS total_belanja FROM fact_orders f JOIN dimcustomer c ON f.id_dimCustomer = c.id_dimCustomer GROUP BY c.customerName ORDER BY total_belanja DESC LIMIT 10;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

nama_pelanggan	jumlah_transaksi	total_unit_dibeli	total_belanja 1
Euro Shopping Channel	259	9327	912294.11
Mini Gifts Distributors Ltd.	180	6366	654858.06
Australian Collectors, Co.	55	1926	200995.41
Muscle Machine Inc	48	1775	197736.94
La Rochelle Gifts	53	1832	180124.90
Dragon Souveniers, Ltd.	43	1524	172989.68
Land of Toys Inc.	49	1631	164069.44
The Sharp Gifts Warehouse	40	1656	160010.27
AV Stores, Co.	51	1778	157807.81
Anna's Decorations, Ltd	46	1469	153996.13

Hasil query KPI Top 10 Pelanggan menunjukkan bahwa data warehouse mampu mengidentifikasi pelanggan paling bernilai bagi perusahaan. Euro Shopping Channel menempati posisi teratas dengan total belanja $912.294,11 dari 259 transaksi, hampir 1,4 kali lipat dibandingkan pelanggan kedua yaitu Mini Gifts Distributors Ltd. dengan $654.858,06. Dua pelanggan teratas ini secara bersama-sama menyumbang sekitar 16% dari total revenue keseluruhan, menunjukkan adanya ketergantungan yang cukup tinggi pada pelanggan kunci. KPI ini bermanfaat bagi tim penjualan dan manajemen untuk merancang strategi retensi pelanggan serta program loyalitas yang diprioritaskan pada pelanggan-pelanggan dengan kontribusi revenue terbesar.

KPI 5: Negara dengan Penjualan Tertinggi

KPI ini mengukur performa penjualan berdasarkan geografi. USA adalah pasar terbesar dengan $3,6 juta (~36% dari total revenue global), jauh melampaui negara lainnya. Eropa secara kolektif (Spain, France, UK, Italy, Finland, Norway, Denmark) menyumbang porsi besar berikutnya, menunjukkan bahwa EMEA adalah wilayah strategis kedua. Insight ini berguna untuk perencanaan ekspansi pasar dan alokasi tenaga penjualan per wilayah.

Query :

```
SELECT
    l.country AS negara,
    l.territory AS wilayah,
    COUNT(f.id_factOrder) AS jumlah_transaksi,
    SUM(f.sales) AS total_revenue
FROM fact_orders f
JOIN dimlocation l ON f.id_dimLocation = l.id_dimLocation
GROUP BY l.country, l.territory
ORDER BY total_revenue DESC;
```

Hasil :

Showing rows 0 - 18 (19 total, Query took 0.0102 seconds.)

```sql
SELECT l.country AS negara, l.territory AS wilayah, COUNT(f.id_factOrder) AS jumlah_transaksi, SUM(f.sales) AS total_revenue FROM fact_orders f JOIN dimlocation l ON f.id_dimLocation = l.id_dimLocation GROUP BY l.country, l.territory ORDER BY total_revenue DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

negara	wilayah	jumlah_transaksi	total_revenue 1
USA	NA	1004	3627982.83
Spain	EMEA	342	1215686.92
France	EMEA	314	1110916.52
Australia	APAC	185	630623.10
UK	EMEA	144	478880.46
Italy	EMEA	113	374674.31
Finland	EMEA	92	329581.91
Norway	EMEA	85	307463.70
Singapore	Japan	79	288488.41
Denmark	EMEA	63	245637.15
Canada	NA	70	224078.56
Germany	EMEA	62	220472.09
Sweden	EMEA	57	210014.21
Austria	EMEA	55	202062.53
Japan	Japan	52	188167.81
Switzerland	EMEA	31	117713.56
Belgium	EMEA	33	108412.62
Philippines	Japan	26	94015.73
Ireland	EMEA	16	57756.43

Hasil query KPI Negara dengan Penjualan Tertinggi menunjukkan bahwa data warehouse berhasil memetakan performa penjualan secara geografis. USA menempati posisi teratas sebagai pasar terbesar dengan total revenue \$3.627.982,83 dari 1.004 transaksi, menyumbang sekitar 36% dari total revenue global. Di urutan berikutnya, negara-negara Eropa seperti Spain, France, UK, dan Italy secara kolektif menunjukkan bahwa wilayah EMEA merupakan kawasan strategis kedua setelah Amerika Serikat. KPI ini bermanfaat bagi manajemen dalam perencanaan ekspansi pasar, penentuan target wilayah prioritas, serta alokasi tenaga penjualan yang lebih efektif berdasarkan potensi revenue di setiap wilayah.

V. KESIMPULAN

Berdasarkan hasil implementasi proyek ETL dan pembangunan Data Warehouse GlobalScale Retail, dapat disimpulkan bahwa seluruh tahapan proyek mulai dari pengambilan data, perancangan skema, hingga implementasi pipeline ETL telah berhasil dilaksanakan dengan baik.

Data Warehouse yang dibangun menggunakan model Star Schema dengan satu tabel fakta dan empat tabel dimensi berhasil menampung dan mengorganisasi 2.823 baris data transaksi dari sumber CSV menjadi struktur yang terorganisir dan siap dianalisis. Proses ETL yang dirancang mampu menangani pembersihan data, konversi format tanggal, deduplikasi, serta pembentukan surrogate key untuk menjaga integritas relasi antar tabel.

Dari hasil analisis lima KPI yang dilakukan terhadap Data Warehouse, diperoleh kesimpulan bahwa perusahaan mengalami pertumbuhan revenue yang positif pada periode 2003–2004, dengan Classic Cars sebagai lini produk paling dominan, Medium Deal sebagai segmen transaksi terbesar, serta Amerika Serikat sebagai pasar utama yang menyumbang lebih dari sepertiga total pendapatan global. Selain itu, teridentifikasi dua pelanggan utama yang memberikan kontribusi signifikan terhadap revenue perusahaan sehingga perlu mendapat perhatian khusus dalam strategi retensi pelanggan.

Dengan demikian, Data Warehouse ini terbukti dapat menjadi fondasi yang kuat untuk mendukung proses analisis data dan pengambilan keputusan bisnis berbasis data secara efisien dan terstruktur.

Link github : https://github.com/nideeuw/Project-UAS_Kelompok-4.git